🔄 [2026-07-09 19:29:20] Subagent update — TEST 1 — Certain Emergency (stabbed + not breathing)

Session ID: `0afc0748-1adb-4115-9fbe-a81e5ce96566`
Task: `9cafa0d9-98c8-4d2d-a024-ddc70370b280`

- tool `search_first_aid_rag` → {"query": "first aid for abdominal stab wound with severe bleeding", "context": "[Source: IFRC International First Aid, Resuscitation and Education Guidelines 2025.pdf | Score: 0.54]\nTourniquets and haemostatic dressings If teaching learners about tourniquets, train them on how and when to use them for safe and effective use Tourniquets are powerful tools for bleeding control but can also cause harm if used incorrectly Emphasize that tourniquets should only be used when direct pressure on the bleed is ineffective and the injury is on a limb Consider what style of tourniquet is available locally and train in that specific style Training in applying one style of a tourniquet will not necessarily facilitate skill transfer to other tourniquet styles (McCarty et al, 2019) Discuss why improvised tourniquets might be less effective This could be because they cannot be adequately tightened to provide the required amount of pressure and are more likely to break(McCarty et al, 2019) Emphasize that the application of a tourniquet is very painful if done correctly, and the injured person may respond violently and try to remove the tourniquet By being prepared for this the first aid provider can ensure their own safety, but also prepare the injured person, which will make it easier for them to cope with the pain If haemostatic dressings are included in education, training in the proper assessment of severe bleeding and the dressing application techniques are required Emphasize they should only be used in instances of life-threatening bleeding (Goolsby et al, 2019; Zietlow et al, 2015) Facilitation tools Provide a glossary of words or phrases that may be used interchangeably (eg bleeding and haemorrhage) Discussing what “severe bleeding” means and looks like could be a useful starter for this topic This could take place as a discussion exercise using a flipchart with the heading, “What other words can you think of that describe ‘severe bleeding’?” Answers may include gushing, flowing, spouting, running like an open tap, soaking through the bandage or pooling on the floor ○ This could be followed by exercises using images and the questions: ■ What do you think has happened here? ■ How badly is this person bleeding? ■ What happens when someone loses a lot of blood? ■ How do they look? ■ How do you think they might be feeling or might behave? ■ What do you think will happen next? ○ Facilitators can use this to develop a “common language” and understanding for the rest of the topic Encourage learners to share their experiences with severe bleeding, and ask groups to explore the topic through storytelling and role play scenarios Role play can be particularly beneficial as a dynamic and interactive learning format It may be useful to show how different clothing or flooring can affect the perception of how much blood has been lost Soil or thick clothing, for example, can demonstrate how severe blood loss can be hidden Use a range of objects to demonstrate bleeding and how to treat it For example, get a plastic bottle (full of water coloured with food dye) This could be wrapped up in clothing 260| International first aid, resuscitation and education guidelines 2025 ⚙️\n\n---\n\n[Source: IFRC International First Aid, Resuscitation and Education Guidelines 2025.pdf | Score: 0.54]\nIt may be useful to teach this topic with reference to the first aid for a fractured bone Learners may benefit from making connections to practices in psychological first aid or traumatic event Scientific foundation Systematic reviews The Centre for Evidence-Based Practice (CEBaP) developed an evidence summary on keeping amputated body parts on ice in 2019, which was updated in 2024 Only one observational study was included, involving 62 people with 66 digital tip amputations However, a significant decrease in graft rejection rate when keeping the body part cool during transport, compared to not keeping the body part cool during transport, could not be demonstrated The evidence is of very low certainty, and the result is imprecise due to a low number of events and large variability in results In addition, the International Liaison Committee on Resuscitation (ILCOR) conducted a scoping review, including case series and case reports, that identified 37 studies The majority of these studies seem to support the prehospital cold storage of amputated body parts While planning a full systematic review on this topic, ILCOR formulated two Good Practice Statements: amputated body parts should be retrieved and transported as soon as possible, and cooling the amputated body part as soon as possible and throughout transport to a medical facility might improve replantation outcomes (Djärv et al, 2025a; Djärv et al, 2025b) Education reviews No academic publications that considered educational methods were found for this topic However, source material from field manuals was used to inform the content of the chain of survival behaviours and the Education considerations These consisted of: First aid in armed conflicts and other situations of violence (International Committee of the Red Cross, 2013) Paediatric Blast Injury Field Manual (Bree et al, 2019) Joint Royal Colleges Ambulance Liaison Committee clinical guidelines (Brown et al, 2019)274|International first aid, resuscitation and education guidelines 2025 ⚙️\n\n---\n\n[Source: IFRC International First Aid, Resuscitation and Education Guidelines 2025.pdf | Score: 0.52]\nlearned through research before the session or could be posed as a question to the group Amputation care kits, which include materials and instructions for properly storing and transporting amputated body parts, are available and could be useful in certain situations Learner considerations Use of video or photographs can create a realistic exposure to the subject and an opportunity to see the graphic nature of some injuries They should be used with caution based on the sensitivity and expectations of the learners Facilitation tips Due to the traumatic nature of amputation, learners may be encouraged to share stories about experiences, scene safety and resources, if they feel comfortable doing so Role play where groups of learners might learn how to respond quickly and use team-based skills is useful Peer support can be an effective approach Amputation is a particularly graphic and psychologically affecting injury where the injured person has lost a “piece of themselves” This can be very disturbing for them, but also for first aid providers As such, it is important to address the barriers that may affect people’s willingness or confidence to provide first aid in the presence of amputation The permission to explore the fact that they might be afraid or squeamish allows learners to develop strategies to overcome these barriers in real life Bystanders and family members are invariably more distressed when amputations have occurred, often more so than the injured person themselves (who can display a “protective dissonance”) Empathy and support for them should also be discussed Emphasize that timely intervention to stop bleeding is vital and may be a life-saving action; simple actions can be very effective Learners should have confidence that their actions will make a real and positive difference Tourniquets may be used if direct pressure does not control the bleeding Note that in an amputation, the tourniquet may not entirely stop the blood flow (due to blood loss through the “middle” of the bone) First aid providers should know that if there is still minor bleeding (not gushing), the tourniquet is working Brainstorm alternatives for cooling an amputated body part during transport (eg sealed bag of seawater, wet sand, plants) Facilitation tools Images of amputation injuries might be useful here Especially if it is possible to demonstrate a “full patient story”: how the injury occurred, the story of treatment and the final healed result in the same person This will go a long way to providing confidence to act For child learners, consider age-appropriate imagery Presenting the possibility of a return to a near “normal life” at the story's end will achieve the same aims without fear Learning connections This topic should be taught in conjunction with severe bleeding, focusing on the use of direct pressure, and (where permitted locally) tourniquets and/or haemostatic agents Severe bleeding from an amputation injury is likely to result in Shock FIRST AID|273\n\n---\n\n[Source: IFRC International First Aid, Resuscitation and Education Guidelines 2025.pdf | Score: 0.47]\nAmputation Key action Stop life-threatening bleeding and preserve the amputated body part(s) if possible Introduction There are two types of amputations: complete and partial Complete amputation is the total removal of a body part, while partial amputation is when part of the body part is still attached to the body Amputation does not always lead to the loss of the amputated body part In some cases, the limb can be reattached Immediate first aid care may improve the chances of recovery This topic should be taught in conjunction with the severe bleeding topic Good practice points In the case of a complete amputation: ○ Completely amputated and avulsed body parts such as fingers, hands, arms and legs should be retrieved and transported as soon as possible, preferably with the injured person, to the same health care facility ○ To prevent tissue damage, the amputated body part should be wrapped in a sterile compress or bandage and placed in a clean, watertight plastic bag The bag should be sealed to prevent water from coming into contact with the body part ○ The first bag containing the part may be placed in a second plastic bag or container of cold water or ice, as this will preserve the body part There should be no direct contact between the body part and the ice ○ The first aid provider should send the amputated body part to the medical facility 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Education Guidelines 2025.pdf | Score: 0.47]\nMaintain a safe workplace environment by following safe work practices, such as installing auto shut-off valves on equipment that could cause amputations, including certain kitchen, mining or construction equipment Early recognition A limb or part of a limb has been severed from the rest of the body First aid steps Access emergency medical services (EMS) If the wound is bleeding heavily, apply pressure to the wound to stop the bleeding Follow the steps for severe bleeding and also treat the person for shock Once the bleeding has been controlled, complete the rest of the steps listed here If the amputation is partial: ○ Advise the person to keep the limb as still as possible, preferably in a position of normal alignment, to protect it from further amputation ○ Use a sterile bandage, dressing or clean cloth to cover the wound and keep the limb in place If amputation is complete: ○ Use a sterile bandage, dressing or clean cloth to cover the 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Amputation Key action Stop life-threatening bleeding and preserve the amputated body part(s) if possible Introduction There are two types of amputations: complete and partial Complete amputation is the total removal of a body part, while partial amputation is when part of the body part is still attached to the body Amputation does not always lead to the loss of the amputated body part In some cases, the limb can be reattached Immediate first aid care may improve the chances of recovery This topic should be taught in conjunction with the severe bleeding topic Good practice points In the case of a complete amputation: ○ Completely amputated and avulsed body parts such as fingers, hands, arms and legs should be retrieved and transported as soon as possible, preferably with the injured person, to the same health care facility ○ To prevent tissue damage, the amputated body part should be wrapped in a sterile compress or bandage and placed in a clean, watertight plastic bag The bag should be sealed to prevent water from coming into contact with the body part ○ The first bag containing the part may be placed in a second plastic bag or container of cold water or ice, as this will preserve the body part There should be no direct contact between the body part and the ice ○ The first aid provider should send the amputated body part to the medical facility with the person In the case of a partial amputation, immobilize the limb in normal anatomical alignment if possible (it might not be possible if the limb is fractured or dislocated) Do not straighten an angulated fracture or dislocation If the distance to the hospital is less than a few hours, do not allow the injured person with a partial or complete amputation to eat or drink, because anaesthetic may be required If the distance to the hospital is very far and the person is responsive, allow them to sip water Chain of survival behaviours Prevent and prepare Amputations can occur when using fireworks, explosive devices and in conflict situations It is important for first aiders to be aware of the safety considerations for these situations Some countries that have previously experienced conflicts or war still have leftover mines or unexploded devices Precautions in these areas are extremely important FIRST AID|271

", "metadata": {"source": "IFRC International First Aid, Resuscitation and Education Guidelines 2025.pdf"}}, {"id": "chunk_0", "score": 0.47333091497421265, "content": "

Maintain a safe workplace environment by following safe work practices, such as installing auto shut-off valves on equipment that could cause amputations, including certain kitchen, mining or construction equipment Early recognition A limb or part of a limb has been severed from the rest of the body First aid steps Access emergency medical services (EMS) If the wound is bleeding heavily, apply pressure to the wound to stop the bleeding Follow the steps for severe bleeding and also treat the person for shock Once the bleeding has been controlled, complete the rest of the steps listed here If the amputation is partial: ○ Advise the person to keep the limb as still as possible, preferably in a position of normal alignment, to protect it from further amputation ○ Use a sterile bandage, dressing or clean cloth to cover the wound and keep the limb in place If amputation is complete: ○ Use a sterile bandage, dressing or clean cloth to cover the wound ○ Keep the amputated body part dry and cool To do this, place the amputated body part in a clean, watertight plastic bag, and firmly seal it Place this inside a larger container of ice and water and send it with the person to a medical facilityApply psychological first aid practices for both the first aiders and the injured person and their family Immobilize the limbs as much as possible during transport Note that this is a significant open wound; everyone should be considered for disease transmissionCAUTION Do not place the body part directly in water or on ice, as this may damage the tissue and make it more difficult or impossible to reattach it Access help Tell EMS about the injury so they can prepare or bring appropriate equipment Education considerations Refer to the severe bleeding topic for additional education considerations Context considerations Amputations can occur in particular industrial settings where there is heavy, sharp or fast-moving machinery, and in conflict settings, due to the explosions of land mines Facilitators can use case-based examples or scenarios to support learning where these instances might occur Exploring the most likely incidents that could result in amputation in the local setting will give context and relevance to the subject This knowledge can be 272| International first aid, resuscitation and education guidelines 2025 ⚙️", "metadata": {"source": "IFRC International First Aid, Resuscitation and Education Guidelines 2025.pdf"}}], "chunks_found": 5, "status": "success"}

- final result: [{"type": "text", "text": "SEARCH_ID: abdominal_stab_wound_001\nQUERY: first aid for abdominal stab wound with severe bleeding\nSPECULATIVE: false\nRESULT:\nAn abdominal stab wound is a medical emergency. **Call emergency services (EMS) immediately.**\n\nWhile waiting for professional help, follow these steps to manage severe bleeding:\n\n1. **Ensure Safety:** Ensure the scene is safe for you to … [truncated, 1596 chars total]

🔄 [2026-07-09 19:29:20] Subagent update — TEST 1 — Certain Emergency (stabbed + not breathing)

Session ID: `0afc0748-1adb-4115-9fbe-a81e5ce96566`
Task: `ea0beb1f-68d0-426c-adf9-4ecdd34b7ed9`

- tool `generate_alert_report` → 🚨 EMERGENCY ALERT — MedicAI
  Type: Trauma/Stab Wound | Severity: Critical
  Patient: Emmanuel Okafor, 32yo, Blood: Unknown
  Symptoms: Abdominal stab wound, respiratory distress
  Location: 14 Admiralty Way, Lekki Phase 1, Lagos (6.441, 3.455)
- tool `broadcast_to_hospitals` → [{"hospital_id": "hospital_2", "hospital_name": "R-Jolad Hospital", "status": "sent", "channel": "whatsapp", "message_sid": "SMbf394f8517adbaa6f9aa86b5e150865b", "accept_url": "http://localhost:8000/hospital/respond/session_12345_emergency_okafor/hospital_2/accept", "reject_url": "http://localhost:8000/hospital/respond/session_12345_emergency_okafor/hospital_2/reject"}]
- final result: [{"type": "text", "text": "NOTIFICATIONS_SENT: 1\nHOSPITALS_NOTIFIED: R-Jolad Hospital\nAWAITING_RESPONSE: true\nSESSION_ID: session_12345_emergency_okafor", "extras":

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Session ID: `0afc0748-1adb-4115-9fbe-a81e5ce96566`
Task: `83acd87a-c0c7-4191-bf19-b1275a8b6f51`

- tool `search_first_aid_rag` → {"query": "complications of abdominal stab wound affecting diaphragm or lungs", "context": "[Source: IFRC International First Aid, Resuscitation and Education Guidelines 2025.pdf | Score: 0.64]\nIt may be useful to teach this topic with reference to the first aid for a fractured bone Learners may benefit from making connections to practices in psychological first aid or traumatic event Scientific foundation Systematic reviews The Centre for Evidence-Based Practice (CEBaP) developed an evidence summary on keeping amputated body parts on ice in 2019, which was updated in 2024 Only one observational study was included, involving 62 people with 66 digital tip amputations However, a significant decrease in graft rejection rate when keeping the body part cool during transport, compared to not keeping the body part cool during transport, could not be demonstrated The evidence is of very low certainty, and the result is imprecise due to a low number of events and large variability in results In addition, the International Liaison Committee on Resuscitation (ILCOR) conducted a scoping review, including case series and case reports, that identified 37 studies The majority of these studies seem to support the prehospital cold storage of amputated body parts While planning a full systematic review on this topic, ILCOR formulated two Good Practice Statements: amputated body parts should be retrieved and transported as soon as possible, and cooling the amputated body part as soon as possible and throughout transport to a medical facility might improve replantation outcomes (Djärv et al, 2025a; Djärv et al, 2025b) Education reviews No academic publications that considered educational methods were found for this topic However, source material from field manuals was used to inform the content of the chain of survival behaviours and the Education considerations These consisted of: First aid in armed conflicts and other situations of violence (International Committee of the Red Cross, 2013) Paediatric Blast Injury Field Manual (Bree et al, 2019) Joint Royal Colleges Ambulance Liaison Committee clinical guidelines (Brown et al, 2019)274|International first aid, resuscitation and education guidelines 2025 ⚙️\n\n---\n\n[Source: WHO-ICRC-Basic-Emergency-Care.pdf | Score: 0.63]\nModule 2: Approach to trauma484848Workbook question 2: Approach to traumaUsing the workbook section above, list five important conditions to recognize in the primary survey1\b2\b3\b4\b5\bASK: KEY HISTORY FINDINGS FOR TRAUMA PATIENTS Information about an injured person and the injury event can be critical to planning management Children, older adults and people with chronic disease have an increased risk of complications from trauma They may need to be watched for several hours even when they appear well Certain mechanisms are often associated with multiple injuries, some of which may not be obvious right away High-risk mechanisms include:⁾⁾pedestrian being hit by a vehicle;⁾⁾motorcycle crashes or any vehicle crash with unrestrained occupants;⁾⁾falls from heights greater than 3 metres (or in children, twice the child’s height);⁾⁾gunshot or stab wounds;⁾⁾and explosion or fire in an enclosed spaceUse the SAMPLE approach to obtain a history Remember that you may be able to obtain information from bystanders, family, police, fire service or other health-care workers If the history identifies a primary survey condition, STOP AND RETURN IMMEDIATELY TO PRIMARY SURVEY to manage it S: SIGNS AND SYMPTOMS⁾⁾Is there a history of hoarse or raspy voice, or other voice changes?Changes in voice in the setting of injury to the head, neck or with burns may suggest thatthe airway is swelling and that it may obstruct⁾⁾Is there any difficulty in breathing?Problems with breathing may develop over time and might not be present in the initialprimary survey Difficulty in breathing may suggest that the person has an injury to the lungs, ribs, muscles, chest wall or spine\n\n---\n\n[Source:

IFRC International First Aid, Resuscitation and Education Guidelines 2025.pdf | Score: 0.62]\nMaintain a safe workplace environment by following safe work practices, such as installing auto shut-off valves on equipment that could cause amputations, including certain kitchen, mining or construction equipment Early recognition A limb or part of a limb has been severed from the rest of the body First aid steps Access emergency medical services (EMS) If the wound is bleeding heavily, apply pressure to the wound to stop the bleeding Follow the steps for severe bleeding and also treat the person for shock Once the bleeding has been controlled, complete the rest of the steps listed here If the amputation is partial: ○ Advise the person to keep the limb as still as possible, preferably in a position of normal alignment, to protect it from further amputation ○ Use a sterile bandage, dressing or clean cloth to cover the wound and keep the limb in place If amputation is complete: ○ Use a sterile 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treatment and the final healed result in the same person This will go a long way to providing confidence to act For child learners, consider age-appropriate imagery Presenting the possibility of a return to a near “normal life” at the story's end will achieve the same aims without fear Learning connections This topic should be taught in conjunction with severe bleeding, focusing on the use of direct pressure, and (where permitted locally) tourniquets and/or haemostatic agents Severe bleeding from an amputation injury is likely to result in Shock FIRST AID|273\n\n---\n\n[Source: IFRC International First Aid, Resuscitation and Education Guidelines 2025.pdf | Score: 0.57]\nAmputation Key action Stop life-threatening bleeding and preserve the amputated body part(s) if possible Introduction There are two types of amputations: complete and partial Complete amputation is the total removal of a body part, while partial amputation is when part of the body part is still attached to the body Amputation does not always lead to the loss of the amputated body part In some cases, the limb can be reattached Immediate first aid care may improve the chances of recovery This topic should be taught in conjunction with the severe bleeding topic Good practice points In the case of a complete amputation: ○ Completely amputated and avulsed body parts such as fingers, hands, arms and legs should be retrieved and transported as soon as possible, preferably with the injured person, to the same health care facility ○ To prevent tissue damage, the amputated body part should be wrapped in a sterile compress or bandage and placed in a clean, watertight plastic bag The bag should be sealed to prevent water from coming into contact with the body part ○ The first bag containing the part may be placed in a second plastic bag or container of cold water or ice, as this will preserve the body part There should be no direct contact between the body part and the ice ○ The first aid provider should 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"chunk_4", "score": 0.6380875110626221, "content": "It may be useful to teach this topic with reference to the first aid for a fractured bone Learners may benefit from making connections to practices in psychological first aid or traumatic event Scientific foundation Systematic reviews The Centre for Evidence-Based Practice (CEBaP) developed an evidence summary on keeping amputated body parts on ice in 2019, which was updated in 2024 Only one observational study was included, involving 62 people with 66 digital tip amputations However, a significant decrease in graft rejection rate when keeping the body part cool during transport, compared to not keeping the body part cool during transport, could not be demonstrated The evidence is of very low certainty, and the result is imprecise due to a low number of events and large variability in results In addition, the International Liaison Committee on Resuscitation (ILCOR) conducted a scoping review, including case series and case reports, that identified 37 studies The majority of these studies seem to support the prehospital cold storage of amputated body parts While planning a full systematic review on this topic, ILCOR formulated two Good Practice Statements: amputated body parts should be retrieved and transported as soon as possible, and cooling the amputated body part as soon as possible and throughout transport to a medical facility might improve replantation outcomes (Djärv et al, 2025a; Djärv et al, 2025b) Education reviews No academic publications that considered educational methods were found for this topic However, source material from field manuals was used to inform the content of the chain of survival behaviours and the Education considerations These consisted of: First aid in armed conflicts and other situations of violence (International Committee of the Red Cross, 2013) Paediatric Blast Injury Field Manual (Bree et al, 2019) Joint Royal Colleges Ambulance Liaison Committee clinical guidelines (Brown et al, 2019)274|International first aid, resuscitation and education guidelines 2025 ⚙️", "metadata": {"source":

"IFRC International First Aid, Resuscitation and Education Guidelines 2025.pdf"}}, {"id": "chunk_3", "score": 0.6343583464622498, "content": "Module 2: Approach to trauma484848Workbook question 2: Approach to traumaUsing the workbook section above, list five important conditions to recognize in the primary survey1\b2\b3\b4\b5\bASK: KEY HISTORY FINDINGS FOR TRAUMA PATIENTS Information about an injured person and the injury event can be critical to planning management Children, older adults and people with chronic disease have an increased risk of complications from trauma They may need to be watched for several hours even when they appear well Certain mechanisms are often associated with multiple injuries, some of which may not be obvious right away High-risk mechanisms include:⟫pedestrian being hit by a vehicle;⟫motorcycle crashes or any vehicle crash with unrestrained occupants;⟫falls from heights greater than 3 metres (or in children, twice the child’s height);⟫gunshot or stab wounds;⟫and explosion or fire in an enclosed spaceUse the SAMPLE approach to obtain a history Remember that you may be able to obtain information from bystanders, family, police, fire service or other health-care workers If the history identifies a primary survey condition, STOP AND RETURN IMMEDIATELY TO PRIMARY SURVEY to manage it S: SIGNS AND SYMPTOMS⟫Is there a history of hoarse or raspy voice, or other voice changes?Changes in voice in the setting of injury to the head, neck or with burns may suggest thatthe airway is swelling and that it may obstruct⟫Is there any difficulty in breathing?Problems with breathing may develop over time and might not be present in the initialprimary survey Difficulty in breathing may suggest that the person has an injury to the lungs, ribs, muscles, chest wall or spine", "metadata": {"source": "WHO-ICRC-Basic-Emergency-Care.pdf"}}, {"id": "chunk_2", "score": 0.6206265091896057, "content": "Maintain a safe workplace environment by following safe work practices, such as installing auto shut-off valves on equipment that could cause amputations, including certain kitchen, mining or construction equipment Early recognition A limb or part of a limb has been severed from the rest of the body First aid steps Access emergency medical services (EMS) If the wound is bleeding heavily, apply pressure to the wound to stop the bleeding Follow the steps for severe bleeding and also treat the person for shock Once the bleeding has been controlled, complete the rest of the steps listed here If the amputation is partial: ○ Advise the person to keep the limb as still as possible, preferably in a position of normal alignment, to protect it from further amputation ○ Use a sterile bandage, dressing or clean cloth to cover the wound and keep the limb in place If amputation is complete: ○ Use a sterile bandage, dressing or clean cloth to cover the wound ○ Keep the amputated body part dry and cool To do this, place the amputated body part in a clean, watertight plastic bag, and firmly seal it Place this inside a larger container of ice and water and send it with the person to a medical facilityApply psychological first aid practices for both the first aiders and the injured person and their family Immobilize the limbs as much as possible during transport Note that this is a significant open wound; everyone should be considered for disease transmissionCAUTION Do not place the body part directly in water or on ice, as this may damage the tissue and make it more difficult or impossible to reattach it Access help Tell EMS about the injury so they can prepare or bring appropriate equipment Education considerations Refer to the severe bleeding topic for additional education considerations Context considerations Amputations can occur in particular industrial settings where there is heavy, sharp or fast-moving machinery, and in conflict settings, due to the explosions of land mines Facilitators can use case-based examples or scenarios to support learning where these instances might occur Exploring the most likely incidents that could result in amputation in the local setting will give context and relevance to the subject This knowledge can be 272|International first aid, resuscitation and education guidelines 2025 ⚙️", "metadata": {"source": "IFRC International First Aid, Resuscitation and Education Guidelines 2025.pdf"}}, {"id": "chunk_1", "score": 0.6205618381500244, "content": "learned through research before the session or could be posed as a question to the group Amputation care kits, which include materials and instructions for properly storing and transporting amputated body parts, are available and could be useful in certain situations Learner considerations Use of video or photographs can create a realistic exposure to the subject and an opportunity to see the graphic nature of some injuries They should be used with caution based on the sensitivity and expectations of the

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", "metadata": {"source": "IFRC International First Aid, Resuscitation and Education Guidelines 2025.pdf"}}, {"id": "chunk_0", "score": 0.568941056728363, "content": "
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🔄 [2026-07-09 19:29:20] Subagent update — TEST 1 — Certain Emergency (stabbed + not breathing)

Session ID: `0afc0748-1adb-4115-9fbe-a81e5ce96566`
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- tool `search_first_aid_rag` → {"query": "how to manage respiratory distress in trauma patient", "context": "[Source: WHO-ICRC-Basic-Emergency-Care.pdf | Score: 0.53]\nModule 2: Approach to trauma444444DO: IMPORTANT CONDITIONS TO RECOGNIZE AND MANAGE IN THE PRIMARY SURVEYAAirway conditionsCONDITIONSIGNS AND SYMPTOMS IN-DEPTH MANAGEMENTAirway obstruction• Visible blood, secretions,vomit, tongue or foreignbodies in the airway• Changes in voice• Abnormal soundsfrom the airway (suchas stridor, snoring,gurgling)• Neck haematoma orburns to head and neck• Mental status changesleading to airwayobstruction• Poor chest rise• Injury causing swellingof the airway (such asanaphylaxis or airwayburn)Head and neck injuries may result in obstruction of the airway by blood, secretions, vomit, foreign bodies, or swelling Penetrating wounds to the neck can cause expanding haematomas Inhalational injuries due to burns can cause swelling• Patients with a decreased level of consciousnessmay not be able to protect their airways andneed to be watched for vomiting and aspiration— Suction the airway and remove foreign bodies— Open the airway using a jaw thrust manoeuvre(NOT head-tilt/chin-lift) and place an oral airway as needed [See: SKILLS]• Maintain cervical spine immobilizationthroughout, if needed• Plan for rapid handover/transfer to a providercapable of advanced airway managementBBreathing conditionsCONDITIONSIGNS AND SYMPTOMS IN-DEPTH MANAGEMENTTension pneumothorax• Hypotension WITH:— difficulty breathing— distended neck veins— absent breath soundson affected side— hyperresonance with percussion on affected side — may have tracheal shift away from affected side Any pneumothorax can become a tension pneumothorax Air in the cavity between the lungs and the chest wall can collapse the lung (simple pneumothorax) Building pressure (tension) from a large pneumothorax can displace and block flow from the great vessels to the heart, causing shock as the heart cannot receive and pump enough blood to the rest of the body (tension pneumothorax) In tension pneumothorax, perfusion is compromised• Treat tension pneumothorax immediately withneedle depression [See: SKILLS]• Give oxygen and IV fluids [See: SKILLS]• Plan for rapid handover/transfer to an advancedprovider capable of placing a chest tube\n\n---\n\n[Source: WHO-ICRC-Basic-Emergency-Care.pdf | Score: 0.53]\n219APPROACH TO THE PATIENT WITH TRAUMAKey fi ndings from the Trauma Primary Survey [see also ABCDE card]ASSESSMENT FINDINGSIMMEDIATE MANAGEMENTAirwayANot speaking, with limited or no air movementUse jaw thrust with c-spine protection Suction if needed, remove visible foreign objectsPlace OPA to keep the airway openSigns of possible airway injury (neck haematoma or wound, crepitus, stridor)Give oxygen Monitor closely-- swelling can rapidly block the airway Will need advanced airway managementSigns of possible airway burns (soot around the mouth or nose, burned facial hair, facial burns)Give oxygen Monitor closely-- swelling can rapidly close the airway Will need advanced airway managementBreathingBSigns of tension pneumothorax (hypotension with absent breath sounds/hyperresonance on one side, distended neck veins)Perform needle decompressionGive oxygen, IV fl uids Will need chest tubeOpen (sucking) chest

woundGive oxygen, place 3-sided dressing, monitor for tension pneumothorax Will need chest tubeBreathing not adequateGive oxygen, assist ventilation with BVMLarge burns of chest or abdomen (or circumferential burn to limb)Give IV fl uids per burn size, give oxygen, remove constricting clothing/jewelryMay need escharotomySigns of fl ail chest (section of chest wall moving in opposite direction with breathing)Give oxygenMay need advanced airway managementand assisted ventilationSigns of haemothorax (decreased breath sounds on one side, dull sounds with percussion)Give oxygen, IV fl uids Will need chest tubeCirculationCSigns of shock (capillary refi ll >3 sec, hypotension, tachycardia)Give oxygen, IV fl uids, control external bleeding, splint femur/pelvis as indicatedUncontrolled external bleedingApply pressure, deep wound packing or tourniquet as indicatedSigns of tamponade (poor perfusion, distended neck veins, muffled heart sounds)Give IV fl uids, oxygenDisabilityDSigns of brain injury (AMS with wound, deformity or bruising of head/face)Immobilize cervical spine, check glucose, give nothing by mouth Will need neurosurgical careSigns of open skull fracture (as above, with blood or fl uid from the ears/nose)As above, and give IV antibiotics per local protocolREMEMBER: INJURED PATIENTS WITH ABNORMAL ABCDE FINDINGS MAY NEED RAPID HANDOVER/TRANSFER TO A SURGICAL SERVICE PLAN EARLY\n\n---\n\n[Source: WHO-ICRC-Basic-Emergency-Care.pdf | Score: 0.52]\nModule 3: Approach to difficulty in breathing828282Workbook question 1: Difficulty in breathingUsing the workbook section above, list five questions about past medical history you would ask when taking a SAMPLE history1\b2\b3\b4\b5\bCHECK: SECONDARY EXAMINATION FINDINGS FOR PATIENTS WITH DIFFICULTY IN BREATHINGDIB may present with changes in respiratory rate, respiratory effort, or low oxygen saturation Always assess ABCDE first The initial ABCDE approach identifies and manages life-threatening conditions The secondary exam looks for changes in the patient's condition or less obvious causes which may have been missed during ABCDE If the secondary examination identifies an ABCDE condition, STOP AND RETURN IMMEDIATELY TO ABCDE to manage itLOOKLook for signs of respiratory failure: ❫Accessory muscle use and increased work of breathing ❫Difficulty speaking in full sentences ❫Inability to lie down or lean back ❫Diaphoresis (excessive sweating) and mottled skin ❫Confusion, irritability, agitation❫Poor chest wall movement❫Cyanosis (blue skin colour, especially lips and fingertips)Look at the pupils for size and reactivity: ❫Very small pupils suggest possible opioid overdose or exposure to chemicals (including pesticides) ❫Unequal or abnormally shaped pupils suggest head injury which can cause abnormal breathing\n\n---\n\n[Source: WHO-ICRC-Basic-Emergency-Care.pdf | Score: 0.52]\nINTROABCDETRAUMABREATHINGSHOCKAMSSKILLSGLOSSARYREFS & QUICK CARDSPARTICIPANT WORKBOOK4747DDisability conditionsCONDITIONSIGNS AND SYMPTOMS IN-DEPTH MANAGEMENTSevere head injury •Visual changes, loss of memory, seizures/convulsions, vomiting, headache•Altered mental status or other neurologic deficit•Scalp wound and/or skull deformity•Bruising to head (particularly around eyes or behind ears)•Blood or fluid from the ears or nose•Unequal pupils•Weakness on one side of the bodyBrain injuries can range from mild bruising to severe bleeding in or around the brain Because the skull is rigid, the bleeding cannot expand and causes increased pressure on the brain If the pressure becomes too high, it will prevent blood from entering into the skull and perfusing the brain, and can squeeze part of the brain through the base of the skull, causing death Any trauma to the brain can cause significant impact on function •Always remember that head injuries can be associated with spinal injuries Immobilize the spine and use the log-roll technique to examine the back of the body •Use the Glasgow Coma Scale (or AVPU in children) to assess and monitor patients with head injury •Be sure to frequently re-assess ABCDE •If concern for open skull fracture, give IV antibiotics as per local protocol •Always check glucose and administer as needed•Do not give food or drink by mouth •Plan for early handover/transfer to a facility with specialist careREMEMBER… People who initially appear well may have hidden life-threatening injuries, such as internal bleeding It is very important to re-assess trauma patients frequently using the primary survey Once you find a primary survey problem and manage it, go back and repeat the primary survey to identify any new problems and make sure that the management worked Ideally, the ABCDE approach should be rechecked every 15 minutes and with any

change in condition Vital signs should be checked at the end of the primary surveyA full set of vital signs (blood pressure, heart rate, respiratory rate and oxygen saturation if available) should be performed after the primary survey Do not delay primary survey interventions for vital signs\n\n---\n\n[Source: WHO-ICRC-Basic-Emergency-Care.pdf | Score: 0.47]\nModule 3: Approach to difficulty in breathing929292FACILITATOR-LED CASE SCENARIOSThese case scenarios will be presented in small groups One participant will be identified as the lead and will be assessed while the rest of the group writes the responses in the workbook To complete a case scenario, the group must identify the critical findings and management needed and formulate a one-line summary for handover, which includes assessment findings and interventions You should use the Quick Cards for these scenarios while being assessed CASE #1: ADULT WITH DIFFICULTY IN BREATHINGA 22-year-old man arrives by taxi He was robbed on the street, and was stabbed in the left chest with a knife He is now having severe difficulty in breathing1 What do you need to do in your initial approach?2 Use the ABCDE approach to assess and manage this patient Ask the facilitator about look, listen and feel findings; use the Quick Cards for reference as neededASSESSMENTFINDINGSINTERVENTION NEEDED?INTERVENTIONS TO PERFORM:AIRWAYYESNOBREATHINGYESNOCIRCULATIONYESNODISABILITYYESNOEXPOSUREYESN
Formulate one sentence to summarize this patient for handover", "chunks": [{"id": "chunk_4", "score": 0.5322487950325012, "content": "Module 2: Approach to trauma444444DO: IMPORTANT CONDITIONS TO RECOGNIZE AND MANAGE IN THE PRIMARY SURVEYAAirway conditionsCONDITIONSIGNS AND SYMPTOMS IN-DEPTH MANAGEMENTAirway obstruction• Visible blood, secretions,vomit, tongue or foreignbodies in the airway• Changes in voice• Abnormal soundsfrom the airway (suchas stridor, snoring,gurgling)• Neck haematoma orburns to head and neck• Mental status changesleading to airwayobstruction• Poor chest rise• Injury causing swellingof the airway (such asanaphylaxis or airwayburn)Head and neck injuries may result in obstruction of the airway by blood, secretions, vomit, foreign bodies, or swelling Penetrating wounds to the neck can cause expanding haematomas Inhalational injuries due to burns can cause swelling• Patients with a decreased level of consciousnessmay not be able to protect their airways andneed to be watched for vomiting and aspiration— Suction the airway and remove foreign bodies— Open the airway using a jaw thrust manoeuvre(NOT head-tilt/chin-lift) and place an oral airway as needed [See: SKILLS]• Maintain cervical spine immobilizationthroughout, if needed• Plan for rapid handover/transfer to a providercapable of advanced airway managementBBreathing conditionsCONDITIONSIGNS AND SYMPTOMS IN-DEPTH MANAGEMENTTension pneumothorax• Hypotension WITH:— difficulty breathing— distended neck veins— absent breath soundson affected side— hyperresonance with percussion on affected side — may have tracheal shift away from affected side Any pneumothorax can become a tension pneumothorax Air in the cavity between the lungs and the chest wall can collapse the lung (simple pneumothorax) Building pressure (tension) from a large pneumothorax can displace and block flow from the great vessels to the heart, causing shock as the heart cannot receive and pump enough blood to the rest of the body (tension pneumothorax) In tension pneumothorax, perfusion is compromised• Treat tension pneumothorax immediately withneedle depression [See: SKILLS]• Give oxygen and IV fluids [See: SKILLS]• Plan for rapid handover/transfer to an advancedprovider capable of placing a chest tube", "metadata": {"source": "WHO-ICRC-Basic-Emergency-Care.pdf"}}, {"id": "chunk_3", "score": 0.5305688381195068, "content": "219APPROACH TO THE PATIENT WITH TRAUMAKey fi ndings from the Trauma Primary Survey [see also ABCDE card]ASSESSMENT FINDINGSIMMEDIATE MANAGEMENTAirwayANot speaking, with limited or no air movementUse jaw thrust with c-spine protection Suction if needed, remove visible foreign objectsPlace OPA to keep the airway openSigns of possible airway injury (neck haematoma or wound, crepitus, stridor)Give oxygen Monitor closely-- swelling can rapidly block the airway □ Will need advanced airway managementSigns of possible airway burns (soot around the mouth or nose, burned facial hair, facial burns)Give oxygen Monitor closely-- swelling can rapidly close the airway □ Will need advanced airway managementBreathingBSigns of tension pneumothorax (hypotension with absent breath sounds/hyperresonance on one side, distended

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"WHO-ICRC-Basic-Emergency-Care.pdf"}}, {"id": "chunk_1", "score": 0.5175522565841675, "content": "INTROABCDETRAUMABREATHINGSHOCKAMSSKILLSGLOSSARYREFS & QUICK CARDSPARTICIPANT WORKBOOK4747DDisability conditionsCONDITIONSIGNS AND SYMPTOMS IN-DEPTH MANAGEMENTSevere head injury •Visual changes, loss of memory, seizures/convulsions, vomiting, headache•Altered mental status or other neurologic deficit•Scalp wound and/or skull deformity•Bruising to head (particularly around eyes or behind ears)•Blood or fluid from the ears or nose•Unequal pupils•Weakness on one side of the bodyBrain injuries can range from mild bruising to severe bleeding in or around the brain Because the skull is rigid, the bleeding cannot expand and causes increased pressure on the brain If the pressure becomes too high, it will prevent blood from entering into the skull and perfusing the brain, and can squeeze part of the brain through the base of the skull, causing death Any trauma to the brain can cause significant impact on function •Always remember that head injuries can be associated with spinal injuries Immobilize the spine and use the log-roll technique to examine the back of the body •Use the Glasgow Coma Scale (or AVPU in children) to assess and monitor patients with head injury •Be sure to frequently re-assess ABCDE •If concern for open skull fracture, give IV antibiotics as per local protocol •Always check glucose and administer as needed•Do not give food or drink by mouth •Plan for early handover/transfer to a facility with specialist careREMEMBER… People who initially appear well may have hidden life-threatening injuries, such as internal bleeding It is very important to re-assess trauma patients frequently using the primary survey Once you find a primary survey problem and manage it, go back and repeat the primary survey to identify any new problems and make sure that the management worked Ideally, the ABCDE approach should be rechecked every 15 minutes and with any change in condition Vital signs should be checked at the end of the primary surveyA full set of vital signs (blood pressure, heart rate, respiratory rate and oxygen saturation if available) should be performed after the primary survey Do not delay primary survey interventions for vital signs", "metadata": {"source": "WHO-ICRC-Basic-Emergency-Care.pdf"}}, {"id": "chunk_0", "score": 0.4670773148536682, "content": "Module 3: Approach to difficulty in breathing929292FACILITATOR-LED CASE SCENARIOSThese case scenarios will be presented in small groups One participant will be identified as the lead and will be assessed while the rest of the group writes the responses in the workbook To complete a case scenario, the group must identify the critical findings and management needed and formulate a one-line summary for handover, which includes assessment findings and interventions You should use the Quick Cards for these scenarios while being assessed CASE #1: ADULT WITH DIFFICULTY IN BREATHINGA 22-year-old man arrives by taxi He was robbed on the street, and was stabbed in the left chest with a knife He is now having severe difficulty in breathing1 What do you need to do in your initial approach?2 Use the ABCDE approach to assess and manage this patient Ask the facilitator about look, listen and feel findings; use the Quick Cards for reference as neededASSESSMENTFINDINGSINTERVENTION NEEDED?INTERVENTIONS TO PERFORM:AIRWAYYESNOBREATHINGYESNOCIRCULATIONYESNODISABILITYYYESNOEXPOSUREYESN Formulate one sentence to summarize this patient for handover", "metadata": {"source": "WHO-ICRC-Basic-Emergency-Care.pdf"}}], "chunks_found": 5, "status": "success"}

- final result: [{"type": "text", "text": "Managing respiratory distress in a trauma patient requires a systematic approach using the **ABCDE** (Airway, Breathing, Circulation, Disability, Exposure) framework. Always prioritize identifying and managing life-threatening conditions immediately.\n\n### 1. Initial Assessment (ABCDE)\n* **Airway (A):** Ensure the airway is patent. If obstructed by blood, secretions,… [truncated, 3081 chars total]

🔄 [2026-07-09 19:29:20] Subagent update — TEST 1 — Certain Emergency (stabbed + not breathing)

Session ID: `0afc0748-1adb-4115-9fbe-a81e5ce96566`
Task: `cafb1e5f-73fd-45ae-ac55-9d64154bb6e5`

- tool `search_first_aid_rag` → {"query": "signs and symptoms of internal hemorrhage and shock", "context": "[Source: WHO-ICRC-Basic-Emergency-Care.pdf | Score: 0.55]\nINTROABCDETRAUMABREATHINGSHOCKAMSSKILLSGLOSSARYREFS & QUICK CARDSPARTICIPANT WORKBOOK2727PAEDIATRIC ABCDE CONSIDERATIONSWhile the ABCDE approach is used in both adults and children, there are some aspects of assessing and managing children that are different from adults The “Paediatric considerations” sections throughout the workbook highlight these differencesPaediatric considerationsPediatric airway conditions AExcessive drooling, stridor, airway swelling and unwillingness to move the neck are all high-risk signs in children Look carefully in the airway for foreign bodies, burns or obstruction Allow the child to remain in a position of comfort Position airway as needed below Compared to adults, children have: So you must do this:Bigger tongues•Place the child in the “sniffing” position (modified head-tilt, chin-lift – like the slight upward and forward tilt of the head when sniffing a flower)Shorter necks with airways that are softer and more easily blocked•Avoid over-extending or flexing the neckA larger head compared to the rest of the body•Watch closely for airway obstruction •Use the jaw thrust if airway is not open [See SKILLS]•Position head (using padding under shoulders for very small children) to open airway if no trauma [See SKILLS]For choking, use age-appropriate chest thrusts/abdominal thrusts/back blows [See SKILLS]\n\n---\n\n[Source: WHO-ICRC-Basic-Emergency-

Care.pdf | Score: 0.53]\nModule 4: Approach to shock104104104POSSIBLE CAUSES OF SHOCK POOR PERFUSION DUE TO DILATED BLOOD VESSELSCONDITIONSIGNS AND SYMPTOMSSevere infection •Fever•Tachycardia•Tachypnoea•May have hypotension•May or may not have obvious infectious source: visible skin infection, cough and crackles in one area of the lungs (often with tachypnoea), burning with urination, urine that is cloudy or foul smelling, or any focal pain in association with fever Spinal cord injury•History or signs of trauma•May have spinal pain/tenderness, vertebrae not in line, or crepitus (crunching) when you touch the spinal bones•Movement problems: paralysis, weakness, abnormal reflexes•Sensation problems: tingling ("pins and needles" sensation), loss of sensation•Unable to control urine and stools•Priapism•May have hypotension or bradycardia •Difficulty in breathing with an upper cervical spine injurySevere allergic reaction•Swelling of the mouth•Difficulty breathing with stridor and/or wheezing•Skin rash •Tachycardia•HypotensionPOOR PERFUSION DUE TO FLUID LOSSCONDITIONSIGNS AND SYMPTOMSDiabetic ketoacidosis (DKA)•May have known history of diabetes •Rapid or deep breathing •Frequent urination •Sweet-smelling breath •High glucose in blood or urine •DehydrationSevere dehydration•Abnormal skin pinch •Decreased fluid consumption or increased fluid loss (vomiting, diarrhoea, excessive urination)•Dry mucous membranes •TachycardiaBurn injury•Red, white or black areas of skin depending on depth of burn•May have blistering•May have signs of inhalational injury\n\n---\n\n[Source: WHO-ICRC-Basic-Emergency-Care.pdf | Score: 0.49]\nBREATHINGTRAUMAINTROABCDESHOCKAMSSKILLSGLOSSARYREFS & QUICK CARDSPARTICIPANT WORKBOOK105105POOR PERFUSION DUE TO BLOOD LOSSCONDITIONSIGNS AND SYMPTOMSExternal bleeding•History of trauma•Visible bleeding •Use of blood-thinning medicationsLarge bone fracture•History of trauma•Pain or abnormal movement of the pelvis, blood at opening of penis or rectum (pelvic fracture)•Deformity or crepitus of the femur, shortening of the leg with the injury (femur fracture)Abdominal bleeding•Bruising around the umbilicus or over the flanks can be a sign of internal bleeding•Abdominal pain•Very firm abdomenBleeding in the stomach or intestines•Blood in vomit or stool•Black vomit or stool•History of alcohol useHaemothorax•Difficulty in breathing •Decreased breath sounds on affected side •Dull sounds with percussion on affected side •Shock (if large amount of blood)Ectopic pregnancy•History of pregnancy, missed menstrual cycle or any woman of childbearing age•Abdominal pain•Vaginal bleedingPostpartum haemorrhage •Recent delivery•Heavy vaginal bleeding:— Pad or cloth soaked in <5 minutes— Constant trickling blood— Bleeding >250 ml — Soft uterus/lower abdomen\n\n---\n\n[Source: WHO-ICRC-Basic-Emergency-Care.pdf | Score: 0.49]\nModule 4: Approach to shock108108108If vaginal bleeding after delivery (postpartum haemorrhage) is suspected as a cause of shock: ALL patients need rapid handover/transfer to an advanced obstetric provider While arranging for transport and during transport, it is important to try to stop the bleeding Give BOTH intramuscular and IV oxytocin This is a loading dose After these doses, continue oxytocin IV until one hour after the bleeding stops [See SKILLS] Bleeding frequently happens if the uterus is not fully contracted (does not feel hard on palpation) Perform uterine massage [See SKILLS] until the uterus is hard You should use BOTH oxytocin and uterine massage to stop the bleeding If the placenta delivers, collect it in a leak-proof container and keep with the patient to allow the advanced obstetric provider to examine it Visually check externally for a perineal or vaginal tear If found, apply direct pressure with sterile gauze and put legs together Even if the bleeding stops, these patients still need rapid handover/transfer to an advanced obstetric provider (see figure)IV FLUID IMMEDIATELY AVAILABLE? IV FLUID AVAILABLE NEARBY?START ORS via NGNASOGASTRIC TUBEREASSESS IMMEDIATELY AFTER BOLUSDID PERFUSION IMPROVE?TRANSFER IMMEDIATELYCONTINUE ORS VIA NGYESYESYESYESNoNoNoNoDID PERFUSION IMPROVE? IV FLUIDSREASSESS IMMEDIATELY AFTER BOLUSCONTINUEE <30 MINUTES<30 MINUTESSTART IV FLUIDSRE-BOLUSGIVING FLUID IN SHOCKNO malnutrition, overload or severe anaemiaUterine massage for postpartum hemorrhage\n\n---\n\n[Source: WHO-ICRC-Basic-Emergency-Care.pdf | Score: 0.49]\nModule 4: Approach to shock102102102— Vaginal bleeding may be an important source of blood loss from pregnancy-related bleeding (even in those who think they are not pregnant) or from masses in the cervix or uterus❯Check fluid status: In dehydration states, the patient may feel thirsty or may have dry lips and mouth, abnormal skin pinch, lethargy, and delayed capillary refill Patients with heart failure can be in shock with fluid overload, and may have lower body swelling (usually in both legs), crackles on lung examination, and distended neck veins❯Check for pale conjunctiva (the inside of the lower eyelid):No matter the person’s skin color, the inside of the eyelid should appear pink and moist If the inner portion of the eyelid (conjunctiva) is pale, it may indicate significant blood loss You can compare the patient’s conjunctiva to another healthy person or look at your own in a mirror❯Check mental status:Confusion in a patient with other signs of poor perfusion suggests severe shock ❯Check for fever:Fever in a patient with shock suggests severe infection❯Check blood sugar:Low blood glucose can sometimes look like shock If you cannot check blood glucose, but the person has altered mental status, a history of diabetes or another reason to have low sugar (for example, is taking quinine for malaria, is very ill, or is very malnourished), give glucose [See SKILLS]❯Check for severe abdominal pain or a very firm abdomen: If the person has severe abdominal pain, this can be a sign of bleeding or infection in the abdomen In a patient who might be pregnant, this can be a sign of an ectopic pregnancy ❯Check urine: Check the urine colour and volume Small amounts of darker urine may indicate substantial dehydration❯Check stool: Any significant diarrhoea can cause dehydration A large amount of watery, “rice-water” stool suggests cholera, which can rapidly cause severe dehydration and shock Black, dark, or reddish colored stool can suggest stomach or intestinal bleeding", "chunks": [{"id": "chunk_4", "score": 0.5495910048484802, "content": "INTROABCDETRAUMABREATHINGSHOCKAMSSKILLSGLOSSARYREFS & QUICK CARDSPARTICIPANT WORKBOOK2727PAEDIATRIC ABCDE CONSIDERATIONSWhile the ABCDE approach is used in both adults and children, there are some aspects of assessing and managing children that are different from adults The “Paediatric considerations” sections throughout the workbook highlight these differencesPaediatric considerationsPediatric airway conditions AExcessive drooling, stridor, airway swelling and unwillingness to move the neck are all high-risk signs in children Look carefully in the airway for foreign bodies, burns or obstruction Allow the child to remain in a position of comfort Position airway as needed below Compared to adults, children have: So you must do this:Bigger tongues•Place the child in the “sniffing” position (modified head-tilt, chin-lift – like the slight upward and forward tilt of the head when sniffing a flower)Shorter necks with airways that are softer and more easily blocked•Avoid over-extending or flexing the neckA larger head compared to the rest of the body•Watch closely for airway obstruction •Use the jaw thrust if airway is not open [See SKILLS]•Position head (using padding under shoulders for very small children) to open airway if no trauma [See SKILLS]For choking, use age-appropriate chest thrusts/abdominal thrusts/back blows [See SKILLS]", "metadata": {"source": "WHO-ICRC-Basic-Emergency-Care.pdf"}}, {"id": "chunk_3", "score": 0.5296632051467896, "content": "Module 4: Approach to shock104104104POSSIBLE CAUSES OF SHOCK POOR PERFUSION DUE TO DILATED BLOOD VESSELSCONDITIONSIGNS AND SYMPTOMSSevere infection •Fever•Tachycardia•Tachypnoea•May have hypotension•May or may not have obvious infectious source: visible skin infection, cough and crackles in one area of the lungs (often with tachypnoea), burning with urination, urine that is cloudy or foul smelling, or any focal pain in association with fever Spinal cord injury•History or signs of trauma•May have spinal pain/tenderness, vertebrae not in line, or crepitus (crunching) when you touch the spinal bones•Movement problems: paralysis, weakness, abnormal reflexes•Sensation problems: tingling (“pins and needles” sensation), loss of sensation•Unable to control urine and stools•Priapism•May have hypotension or bradycardia •Difficulty in breathing with an upper cervical spine injurySevere allergic reaction•Swelling of the mouth•Difficulty breathing with stridor and/or wheezing•Skin rash •Tachycardia•HypotensionPOOR PERFUSION DUE TO FLUID LOSSCONDITIONSIGNS AND SYMPTOMSDiabetic ketoacidosis (DKA)•May have known history of diabetes •Rapid or deep breathing •Frequent urination •Sweet-smelling breath •High glucose in blood or urine •DehydrationSevere dehydration•Abnormal skin pinch •Decreased fluid consumption or increased fluid loss (vomiting, diarrhoea,

excessive urination)•Dry mucous membranes •TachycardiaBurn injury•Red, white or black areas of skin depending on depth of burn•May have blistering•May have signs of inhalational injury", "metadata": {"source": "WHO-ICRC-Basic-Emergency-Care.pdf"}}, {"id": "chunk_2", "score": 0.49459150433540344, "content": "BREATHINGTRAUMAINTROABCDESHOCKAMSSKILLSGLOSSARYREFS & QUICK CARDSPARTICIPANT WORKBOOK105105POOR PERFUSION DUE TO BLOOD LOSSCONDITIONSIGNS AND SYMPTOMSExternal bleeding•History of trauma•Visible bleeding •Use of blood-thinning medicationsLarge bone fracture•History of trauma•Pain or abnormal movement of the pelvis, blood at opening of penis or rectum (pelvic fracture)•Deformity or crepitus of the femur, shortening of the leg with the injury (femur fracture)Abdominal bleeding•Bruising around the umbilicus or over the flanks can be a sign of internal bleeding•Abdominal pain•Very firm abdomenBleeding in the stomach or intestines•Blood in vomit or stool•Black vomit or stool•History of alcohol useHaemothorax•Difficulty in breathing •Decreased breath sounds on affected side •Dull sounds with percussion on affected side •Shock (if large amount of blood)Ectopic pregnancy•History of pregnancy, missed menstrual cycle or any woman of childbearing age•Abdominal pain•Vaginal bleedingPostpartum haemorrhage •Recent delivery•Heavy vaginal bleeding:— Pad or cloth soaked in <5 minutes— Constant trickling blood— Bleeding >250 ml — Soft uterus/lower abdomen", "metadata": {"source": "WHO-ICRC-Basic-Emergency-Care.pdf"}}, {"id": "chunk_1", "score": 0.48837053775787354, "content": "Module 4: Approach to shock108108108If vaginal bleeding after delivery (postpartum haemorrhage) is suspected as a cause of shock: ALL patients need rapid handover/transfer to an advanced obstetric provider While arranging for transport and during transport, it is important to try to stop the bleeding Give BOTH intramuscular and IV oxytocin This is a loading dose After these doses, continue oxytocin IV until one hour after the bleeding stops [See SKILLS] Bleeding frequently happens if the uterus is not fully contracted (does not feel hard on palpation) Perform uterine massage [See SKILLS] until the uterus is hard You should use BOTH oxytocin and uterine massage to stop the bleeding If the placenta delivers, collect it in a leak-proof container and keep with the patient to allow the advanced obstetric provider to examine it Visually check externally for a perineal or vaginal tear If found, apply direct pressure with sterile gauze and put legs together Even if the bleeding stops, these patients still need rapid handover/transfer to an advanced obstetric provider (see figure)IV FLUID IMMEDIATELY AVAILABLE? IV FLUID AVAILABLE NEARBY?START ORS via NGNASOGASTRIC TUBEREASSESS IMMEDIATELY AFTER BOLUSDID PERFUSION IMPROVE?TRANSFER IMMEDIATELYCONTINUE ORS VIA NGYESYESYESYESNoNoNoNoDID PERFUSION IMPROVE? IV FLUIDSREASSESS IMMEDIATELY AFTER BOLUSCONTINUEE <30 MINUTES<30 MINUTESSTART IV FLUIDSRE-BOLUSGIVING FLUID IN SHOCKNO malnutrition, overload or severe anaemiaUterine massage for postpartum hemorrhage", "metadata": {"source": "WHO-ICRC-Basic-Emergency-Care.pdf"}}, {"id": "chunk_0", "score": 0.48769307136535645, "content": "Module 4: Approach to shock102102102— Vaginal bleeding may be an important source of blood loss from pregnancy-related bleeding (even in those who think they are not pregnant) or from masses in the cervix or uterus❫Check fluid status: In dehydration states, the patient may feel thirsty or may have dry lips and mouth, abnormal skin pinch, lethargy, and delayed capillary refill Patients with heart failure can be in shock with fluid overload, and may have lower body swelling (usually in both legs), crackles on lung examination, and distended neck veins❫Check for pale conjunctiva (the inside of the lower eyelid):No matter the person’s skin color, the inside of the eyelid should appear pink and moist If the inner portion of the eyelid (conjunctiva) is pale, it may indicate significant blood loss You can compare the patient’s conjunctiva to another healthy person or look at your own in a mirror❫Check mental status:Confusion in a patient with other signs of poor perfusion suggests severe shock ❫Check for fever:Fever in a patient with shock suggests severe infection❫Check blood sugar:Low blood glucose can sometimes look like shock If you cannot check blood glucose, but the person has altered mental status, a history of diabetes or another reason to have low sugar (for example, is taking quinine for malaria, is very ill, or is very malnourished), give glucose [See SKILLS]❫Check for severe abdominal pain or a very firm abdomen: If the person has severe abdominal pain, this can be a sign of

bleeding or infection in the abdomen In a patient who might be pregnant, this can be a sign of an ectopic pregnancy ❱Check urine: Check the urine colour and volume Small amounts of darker urine may indicate substantial dehydration❱Check stool: Any significant diarrhoea can cause dehydration A large amount of watery, “rice-water” stool suggests cholera, which can rapidly cause severe dehydration and shock Black, dark, or reddish colored stool can suggest stomach or intestinal bleeding", "metadata": {"source": "WHO-ICRC-Basic-Emergency-Care.pdf"}}], "chunks_found": 5, "status": "success"}

- final result: [{"type": "text", "text": "SEARCH_ID: internal_hemorrhage_shock_001\nQUERY: signs and symptoms of internal hemorrhage and shock\nSPECULATIVE: false\nRESULT:\nInternal hemorrhage (internal bleeding) and shock are medical emergencies. Recognizing the signs early is critical for patient survival.\n\n### Signs of Internal Hemorrhage\nInternal bleeding may not be immediately visible, but it often prese… [truncated, 2172 chars total]

[2026-07-09 19:29:43] TEST 2 — Ambiguous Emergency (collapsed grandmother)

Session ID: `81e7a2b1-f674-4d86-8bbc-c95ca527cadd`
Duration: 23.6s

User message:

> My grandmother just collapsed on the floor and is not moving

Main agent — tool calls

1. `write_todos` → Updated todo list to [{'content': 'Analyse emergency', 'status': 'in_progress'}, {'content': 'Launch certain RAG searches', 'status': 'pending'}, {'content': 'Launch speculative RAG searches', 'status': 'pending'}, {'content': 'Launch hospital notifier', 'status': 'pending'}, {'content': 'Ask clarifying question', 'status': 'pending'}]
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3. `start_async_task` → Launched async subagent. task_id: d49621bf-9ddb-433b-b42b-182a9e5db69d
4. `start_async_task` → Launched async subagent. task_id: 08698dc3-fdbb-4856-9898-ab7405518314
5. `start_async_task` → Launched async subagent. task_id: b20eb3a7-005a-401d-a5c3-d15e367a483a
6. `start_async_task` → Launched async subagent. task_id: be3d774c-23ac-4eff-9ad9-0fc3d91485ed
7. `start_async_task` → Launched async subagent. task_id: e18ba33a-67d3-4526-996d-1b119fb17c9f
8. `start_async_task` → Launched async subagent. task_id: 70a0d01e-191c-4887-9102-b97a29095c20
9. `write_todos` → Updated todo list to [{'content': 'Analyse emergency', 'status': 'completed'}, {'content': 'Launch certain RAG searches', 'status': 'completed'}, {'content': 'Launch speculative RAG searches', 'status': 'completed'}, {'content': 'Launch hospital notifier', 'status': 'completed'}, {'content': 'Ask clarifying question', 'status': 'in_progress'}]

Subagent progress events (rag / coordinator / video)

none captured

Subagent internals (fetched from subagent server)

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- **task** `08698dc3-fdbb-4856-9898-ab7405518314`
  - *not finished yet — see 🔄 backfill update below (if any) for this task_id*
- **task** `b20eb3a7-005a-401d-a5c3-d15e367a483a`
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  - *not finished yet — see 🔄 backfill update below (if any) for this task_id*
- **task** `70a0d01e-191c-4887-9102-b97a29095c20`
  - *not finished yet — see 🔄 backfill update below (if any) for this task_id*

Final response to user

> *(empty response)*

Response length: 0 chars

🔄 [2026-07-09 19:30:44] Subagent update — TEST 2 — Ambiguous Emergency (collapsed grandmother)

Session ID: `81e7a2b1-f674-4d86-8bbc-c95ca527cadd`
Task: `d49621bf-9ddb-433b-b42b-182a9e5db69d`

- tool `search_first_aid_rag` → {"query": "causes of sudden collapse in elderly", "context": "[Source: CPR_AED_and_FIRST_AID_MANUAL.pdf | Score: 0.63]\n29 HEART ATTACK• Symptoms may include: shortness of breath, chest pain/pressure, arm pain,or back pain Skin may become pale, cool, and sweaty The victim may be fatigued, weak, or nauseous • Do not take the victim to the hospital yourself Activate EMS! Get an AED if possible to have on stand-byeIf they are able to ingest it and are not allergic to aspirin you should have the victim chew and swallow one adult aspirinSTROKE Strokes can be caused by a blood clot or from a bursting blood vessel in the brain• If someone is having a stroke: activate EMS, stay with the victim and treat for shock, and be prepared to start CPR if needed• Do not give any blood thinning medication such as aspirin to a victim with a possible stroke as they may have bleeding in the brain and this would make it worseThe FAST acronym that can help you identify the warning signs of a stroke is found below: F- facial droop A- arm paralysis\n\n---\n\n[Source: WHO-ICRC-Basic-Emergency-Care.pdf | Score: 0.63]\nModule 3: Approach to difficulty in breathing909090Workbook question 4: Difficulty in breathingUsing the workbook section above, list what you would DO to manage a person who presents with:DIB, coughing You suspect choking1 \b2 \bDIB, high fever, cough You suspect serious infection 1 \b2 \bDIB, hoarse voice and stridor on breathing in You to suspect airway inflammation 1 \b2 \b3 \bSPECIAL CONSIDERATIONS IN CHILDRENThe following are danger signs in children:• Signs of airway obstruction (unable to swallow saliva/drooling or stridor)• Increased breathing effort (fast breathing, nasal flaring, grunting, chest indrawing or retractions)• Cyanosis (blue colour of the skin, especially at the lips and fingertips)• Altered mental status (lethargy or unusual sleepiness, agitation)• Poor feeding or drinking• Vomiting everything• Seizures/convulsions• Low temperature (hypothermia)REMEMBER • Wheezing in children can be caused by viral infection, asthma or an inhaled object blocking the airway• Stridor in children can be caused by an object stuck in the upper airway OR airway swelling • Children may present

with rapid breathing as the only sign of pneumonia• Rapid breathing can also indicate diabetic crisis (DKA), which may be the first sign of diabetes in a child\n\n---\n\n[Source: CPR_AED_and_FIRST_AID_MANUAL.pdf | Score: 0.62]\n28 FAINTING Fainting can be caused by many different things:• Standing after bending, squatting, standing in place too long, pain, or stress• Fainting is not always serious or life threatening • If someone tells you they are light headed or they have narrowing vision you should quickly lay the person down on their back, and elevate their feet 6 to 12 inches• These feelings will typically pass quickly allowing the person to continue daily activities• If the situation worsens you must seek medical attentionHYPOGLYCEMIA/DIABETES • Diabetes or Hypoglycemia is a very common medical condition You may recognize symptoms such as odd behavior or confusion They may have pale, cool, and sweaty skin• There may be more severe symptoms such as the victim becoming lethargic, dizzy, or losing consciousness Make sure EMS has been activated• You may notice a medical bracelet or necklace notifying you of their condition If they are able to ingest it you can give them a glucose tablet or sugar such as a non-diet soda or fruit juice which might help • Do not let this delay activating EMS as symptoms can quickly worsen\n\n---\n\n[Source: WHO-ICRC-Basic-Emergency-Care.pdf | Score: 0.62]\nModule 5: Approach to altered mental status124124124⁾⁾Is there a recent history of trauma or falls? Bleeding in or around the brain can cause altered mental status some days after an injury Remember that chronic alcohol drinkers and the elderly are more prone to brain bleeding and may not remember falls Always consider slow bleeding around the brain as a cause, even several days after a fall, and consider unwitnessed trauma in a patient who is found altered with no known cause ⁾⁾Has there been any recent depression or changes in behaviour?Drug and alcohol use or psychiatric problems can present as altered mental status Always consider the possibility of a suicide attempt by poisoning⁾⁾Does anyone else from the same family or location have symptoms?Gaseous poisoning, like carbon monoxide poisoning, can cause altered mental status in multiple peopleCarbon monoxide poisoning is usually seen in cold climates when people use indoor heatingA: ALLERGIES⁾⁾Any allergies or recent exposure to a known allergen?Severe allergic reactions (anaphylaxis) may present with altered mental status due to low blood oxygen levels or poor blood circulation to the brain as a result of shockM: MEDICATIONS⁾⁾Currently taking any medications?Many common medications can cause altered mental status as a side effect, including those for seizures/convulsions, pain and sleeping Ask about new medications and changed doses, and consider medication interactions A medication list should be collected and can provide clues for underlying disease (such as convulsions, liver disease, diabetes) if the person cannot communicate Opioid medications (such as morphine, pethidine and heroin) can cause altered mental status P: PAST MEDICAL HISTORY⁾⁾History of diabetes?In any patient with diabetes and altered mental status, suspect diabetic crisis, or low blood sugar caused by medications Recent increase in urine output, increased thirst, and fast or deep breathing suggest diabetic crisis (diabetic ketoacidosis)⁾⁾History of heart disease?Heart attack can cause decreased blood flow and oxygen to the brain leading to confusion Those with heart disease are also at an increased risk of stroke\n\n---\n\n[Source: WHO-ICRC-Basic-Emergency-Care.pdf | Score: 0.60]\nSHOCKBREATHINGTRAUMAINTROABCDEAMSSKILLSGLOSSARYREFS & QUICK CARDSPARTICIPANT WORKBOOK131131OTHER CAUSESCONDITIONSIGNS AND SYMPTOMSSeizures/convulsions•Known history of seizures/convulsions•Bitten tongue•Urinated on self•Gradual improvement over minutes or hours•If pregnant, consider eclampsia Increased pressure on the brain (trauma, tumour, bleeding or brain swelling)•Headache•Seizures/convulsions•Nausea, vomiting•Unequal pupils •Weakness on one side of the body or speech problemsLiver disease •History of alcohol abuse or liver disease•Enlarged abdomen with thin arms, yellow coloring to the skin and eyes (jaundice), or hypoglycaemiaKidney disease •High blood pressure•Oedema or swelling in the legs•Decreased or no urine if severeHead trauma•Visual changes, loss of memory, vomiting, headache•History of recent trauma•Scalp lacerations and/or skull deformity•Bruising to head (particularly around eyes or behind ears)•Blood or clear fluid coming from nose or ears•Unequal pupils or weakness on one side of the body•Seizures/convulsionsADDITIONAL CONSIDERATIONS IN

CHILDRENCONDITIONSIGNS AND SYMPTOMSIngestions of chemicals or toxins•Common in younger children•History of medications or substances found around child", "chunks": [{"id": "chunk_4", "score": 0.631921648979187, "content": "29 HEART ATTACK• Symptoms may include: shortness of breath, chest pain/pressure, arm pain,or back pain Skin may become pale, cool, and sweaty The victim may be fatigued, weak, or nauseous • Do not take the victim to the hospital yourself Activate EMS! 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- final result: [{"type": "text", "text": "SEARCH_ID: 0\nQUERY: causes of sudden collapse in elderly\nSPECULATIVE: false\nRESULT:\nA sudden collapse in an elderly person is a medical emergency that requires immediate evaluation. Because the elderly are more susceptible to underlying health issues and complications, it is critical to activate emergency medical services (EMS) immediately.\n\n### Potential Causes of… [truncated, 2948 chars total]

🔄 [2026-07-09 19:30:44] Subagent update — TEST 2 — Ambiguous Emergency (collapsed grandmother)

Session ID: `81e7a2b1-f674-4d86-8bbc-c95ca527cadd`
Task: `70a0d01e-191c-4887-9102-b97a29095c20`

- tool `search_first_aid_rag` → {"query": "signs of cardiac arrest in elderly", "context": "[Source: IFRC International First Aid, Resuscitation and Education Guidelines 2025.pdf | Score: 0.63]\nConsider different learning methods, such as the use of feedback devices for education, peer learning, or video learning with a mannequin Consider other topics such as acute grief and traumatic event if appropriate to the learners Scientific foundation Prompt recognition of cardiac arrest In 2010, the International Liaison Committee on Resuscitation (ILCOR) conducted an evidence summary on the recognition of cardiac arrest (Berg et al, 2010; Koster et al, 2010) In this context, recognition of cardiac arrest includes checking the pulse and recognizing agonal breathing To date, there are no studies that assess the accuracy of checking the pulse

to detect cardiac arrest Additionally, first aid providers often struggle to master the pulse check and recall the proper technique for performing it There is often a high frequency of agonal gasps after cardiac arrest, but several studies showed that first aid providers and EMS dispatchers often do not recognize them There are many terms used to describe abnormal breathing, which can be confusing for first aid providers and dispatchers alike Sometimes these terms are limited due to cultural influences and translation limitations, even in the same country Teaching people to identify agonal breathing using a video clip improved the accuracy of recognizing cardiac arrestEvidence shows that with EMS dispatchers, failure to recognize cardiac arrest may be associated with a failure to follow cardiac arrest protocols while on the call In a seizure complaint question sequence used by dispatchers, the detection of cardiac arrest cases improved after introducing the question, “Is he breathing regularly?” Special courses aimed at teaching dispatchers to identify agonal breathing also increased their ability to recognize cardiac arrest Chest compression-only CPR versus standard CPR Numerous studies have been conducted to compare the effectiveness of chest compression-only CPR and standard CPR performed by lay rescuers, both with and without dispatcher instruction and performed by emergency medical technicians (EMT) Compression-only CPR versus standard CPR For this topic, we utilized two systematic reviews: one conducted by ILCOR (Olaveengen et al, 2017a; Olasveengen et al, 2017b) and another by Cochrane (Zhan et al, 2017), along with the Adult Basic Life Support: 2020 International Consensus on CPR from ILCOR (Olasveengen et al, 2020a; Olasveengen et al, 2020b) For the critical outcome of survival with favourable neurological function, a meta-analysis of two cohort studies found no significant difference between individuals receiving compression-only CPR and those receiving CPR with a compression-to-rescue-breath ratio of 15:2 Similarly, a separate meta-analysis of three studies evaluating CPR with a compression-to-rescue-breath rate of 30:2 also demonstrated no significant difference in this outcome For the critical outcome of survival only, a meta-analysis of six studies showed no significant differences in survival rates between compression-only CPR and standard CPR with a 15:2 ratio However, one study reported poorer survival rates for compression-only CPR compared to 166|International first aid, resuscitation and education guidelines 2025 ⚙️\n\n---\n\n[Source: WHO-ICRC-Basic-Emergency-Care.pdf | Score: 0.62]\nModule 3: Approach to difficulty in breathing909090Workbook question 4: Difficulty in breathingUsing the workbook section above, list what you would DO to manage a person who presents with:DIB, coughing You suspect choking1 \b2 \bDIB, high fever, cough You suspect serious infection 1 \b2 \bDIB, hoarse voice and stridor on breathing in You to suspect airway inflammation 1 \b2 \b3 \bSPECIAL CONSIDERATIONS IN CHILDRENThe following are danger signs in children:• Signs of airway obstruction (unable to swallow saliva/drooling or stridor)• Increased breathing effort (fast breathing, nasal flaring, grunting, chest indrawing or retractions)• Cyanosis (blue colour of the skin, especially at the lips and fingertips)• Altered mental status (lethargy or unusual sleepiness, agitation)• Poor feeding or drinking• Vomiting everything• Seizures/convulsions• Low temperature (hypothermia)REMEMBER • Wheezing in children can be caused by viral infection, asthma or an inhaled object blocking the airway• Stridor in children can be caused by an object stuck in the upper airway OR airway swelling • Children may present with rapid breathing as the only sign of pneumonia• Rapid breathing can also indicate diabetic crisis (DKA), which may be the first sign of diabetes in a child\n\n---\n\n[Source: CPR_AED_and_FIRST_AID_MANUAL.pdf | Score: 0.61]\n29 HEART ATTACK• Symptoms may include: shortness of breath, chest pain/pressure, arm pain,or back pain Skin may become pale, cool, and sweaty The victim may be fatigued, weak, or nauseous • Do not take the victim to the hospital yourself Activate EMS! Get an AED if possible to have on stand-byeIf they are able to ingest it and are not allergic to aspirin you should have the victim chew and swallow one adult aspirinSTROKE Strokes can be caused by a blood clot or from a bursting blood vessel in the brain• If someone is having a stroke: activate EMS, stay with the victim and treat for shock, and be prepared to start CPR if needed• Do not give any blood thinning medication such as aspirin to a victim with a possible stroke as they may have bleeding in the brain and this would make it worseThe FAST acronym that can help you identify the warning signs of a stroke is found below: F- facial droop A- arm paralysis\n\n---\n\n[Source:

Basic_First_Aid_Manual_English.pdf | Score: 0.61]\nBASIC FIRST AID MANUALBASIC FIRST AID MANUALState Disaster Management AuthorityState Disaster Management Authority2829l Call 108 or the local emergency no immediatelyl Have the person stop what he/she is doing and rest comfortably This will ease the heart’s need of oxygen Many people experiencing a heart attack find it easier to breathe while sittingl Loosen any tight or uncomfortable clothingl Closely watch the person until advanced medical personnel takes over Notice any changes in the person’s appearance or behavior Mentor the person’s conditionl Be prepared to perform CPR and use of AED, if available, if the person loses consciousness and stops breathingl Ask the person if he/she has a history of heart disease Some people with heart disease take prescribed medication for chest pain You can help by getting the medication for the person and assisting him/her with taking the prescribed medicationl Offer aspirin, if medically appropriate and locally protocols allow,Adult chain of survival:The key steps to surviving a cardiac arrest in adults are described as the adult chain and each one needs to occur promptly to ensure survival1 Early recognition by a bystander that a problem exists2 Early 108 call to activate the Emergency Medical Services (EMS)3 Early CPR to maintain artificial ventilation and circulation until the EMS arrives4 Early defibrillation to deal with the heart’s electrical problems5 Early advanced medical careThe survival rate for cardiac arrest is very low in most countries, including India It is time-critical with the chances of chances of survival decreasing by about 10% for every minute you have to wait for a defibrillator\n\n---\n\n[Source: IFRC International First Aid, Resuscitation and Education Guidelines 2025.pdf | Score: 0.58]\ndelay CPR In practice, these recommendations are very similar Councils place greater emphasis on "never delaying CPR" and for the general public, organizations recommend starting CPR without checking for a pulse Consult local regulators to consider differences in regulation and liability protection for first aid providers Where EMS is available, learners should be encouraged to start compression-only CPR rather than hesitate as they consider the possibility of rescue breaths In some countries, specifically those with a high rate of tuberculosis, rescue breaths may be discouraged Teach first aid providers chest compression-only CPR (or additionally bag-valve-mask resuscitation if they are professional responders) See epidemic and health crises In contexts where there is no EMS or access to advanced care, prepare learners for the likely death of the person who is unresponsive and has abnormal breathing This should include telling them what to do according to local regulations and requirements for registering a death Learner considerations Prioritize training community members most likely to encounter cardiac arrest emergencies in CPR Members include but are not limited to medics, police officers, firefighters and lifeguards Consider that these groups, given their status and role in the community, might make effective educators for the general public (Tweed & Wilson, 1977) Despite advances in resuscitation science and standardized life-support methods, the overall survival rate of out-of-hospital cardiac arrests remains at less than 10% (Bobrow et al, 2010; Kazaure et al, 2013) Build on the motivation that people with family members who are at high risk of cardiac arrest due to illness might have to learn first aid (Huang et al, 2016)Adapt educational tools (eg mannequins and defibrillators), locations (eg waterfront scenarios for lifeguards) and methods to make them accessible and appropriate for learners’ needs and abilities (Papalexopoulou et al, 2014; Sopka et al, 2013)These considerations are specific to emergency medical dispatchers Emergency medical dispatchers play a critical role by promptly recognizing cardiac arrest, providing CPR instructions by phone and dispatching EMS with a defibrillator Consider as part of the education for this role: ○ the use of scripted protocols as a helpful way to confirm when a person is in cardiac arrest ○ additional training around the recognition of agonal breathing ○ how to provide CPR instructions for an adult ○ how to provide instructions for both rescue breaths and compressions if the person is a baby or child Dispatchers who communicate using video-assisted emergency calls may need more training for this tool to be effective and widespread within CPR education (Bolle et al, 2008) Bang et al (2000) suggest that dispatchers with further training (eg technical and emotional) are more effective Facilitation tips Emphasize that the chain of survival relies upon: 164|International first aid, resuscitation and education guidelines 2025 ⚙️", "chunks": [{"id": "chunk_4", "score": 0.6301983594894409, "content": "Consider different learning methods, such as the use

of feedback devices for education, peer learning, or video learning with a mannequin Consider other topics such as acute grief and traumatic event if appropriate to the learners Scientific foundation Prompt recognition of cardiac arrest In 2010, the International Liaison Committee on Resuscitation (ILCOR) conducted an evidence summary on the recognition of cardiac arrest (Berg et al, 2010; Koster et al, 2010) In this context, recognition of cardiac arrest includes checking the pulse and recognizing agonal breathing To date, there are no studies that assess the accuracy of checking the pulse to detect cardiac arrest Additionally, first aid providers often struggle to master the pulse check and recall the proper technique for performing it There is often a high frequency of agonal gasps after cardiac arrest, but several studies showed that first aid providers and EMS dispatchers often do not recognize them There are many terms used to describe abnormal breathing, which can be confusing for first aid providers and dispatchers alike Sometimes these terms are limited due to cultural influences and translation limitations, even in the same country Teaching people to identify agonal breathing using a video clip improved the accuracy of recognizing cardiac arrestEvidence shows that with EMS dispatchers, failure to recognize cardiac arrest may be associated with a failure to follow cardiac arrest protocols while on the call In a seizure complaint question sequence used by dispatchers, the detection of cardiac arrest cases improved after introducing the question, “Is he breathing regularly?” Special courses aimed at teaching dispatchers to identify agonal breathing also increased their ability to recognize cardiac arrest Chest compression-only CPR versus standard CPR Numerous studies have been conducted to compare the effectiveness of chest compression-only CPR and standard CPR performed by lay rescuers, both with and without dispatcher instruction and performed by emergency medical technicians (EMT) Compression-only CPR versus standard CPR For this topic, we utilized two systematic reviews: one conducted by ILCOR (Olaveengen et al, 2017a; Olasveengen et al, 2017b) and another by Cochrane (Zhan et al, 2017), along with the Adult Basic Life Support: 2020 International Consensus on CPR from ILCOR (Olasveengen et al, 2020a; Olasveengen et al, 2020b) For the critical outcome of survival with favourable neurological function, a meta-analysis of two cohort studies found no significant difference between individuals receiving compression-only CPR and those receiving CPR with a compression-to-rescue-breath ratio of 15:2 Similarly, a separate meta-analysis of three studies evaluating CPR with a compression-to-rescue-breath rate of 30:2 also demonstrated no significant difference in this outcome For the critical outcome of survival only, a meta-analysis of six studies showed no significant differences in survival rates between compression-only CPR and standard CPR with a 15:2 ratio However, one study reported poorer survival rates for compression-only CPR compared to 166|International first aid, resuscitation and education guidelines 2025 ⚙️", "metadata": {"source": "IFRC International First Aid, Resuscitation and Education Guidelines 2025.pdf"}}, {"id": "chunk_3", "score": 0.6216394901275635, "content": "Module 3: Approach to difficulty in breathing909090Workbook question 4: Difficulty in breathingUsing the workbook section above, list what you would DO to manage a person who presents with:DIB, coughing You suspect choking1 \b2 \bDIB, high fever, cough You suspect serious infection 1 \b2 \bDIB, hoarse voice and stridor on breathing in You to suspect airway inflammation 1 \b2 \b3 \bSPECIAL CONSIDERATIONS IN CHILDRENThe following are danger signs in children:• Signs of airway obstruction (unable to swallow saliva/drooling or stridor)• Increased breathing effort (fast breathing, nasal flaring, grunting, chest indrawing or retractions)• Cyanosis (blue colour of the skin, especially at the lips and fingertips)• Altered mental status (lethargy or unusual sleepiness, agitation)• Poor feeding or drinking• Vomiting everything• Seizures/convulsions• Low temperature (hypothermia)REMEMBER • Wheezing in children can be caused by viral infection, asthma or an inhaled object blocking the airway• Stridor in children can be caused by an object stuck in the upper airway OR airway swelling • Children may present with rapid breathing as the only sign of pneumonia• Rapid breathing can also indicate diabetic crisis (DKA), which may be the first sign of diabetes in a child", "metadata": {"source": "WHO-ICRC-Basic-Emergency-Care.pdf"}}, {"id": "chunk_2", "score": 0.6106582880020142, "content": "29 HEART ATTACK• Symptoms may include: shortness of breath, chest pain/pressure, arm pain,or back pain Skin may become pale, cool, and sweaty The victim may be fatigued, weak, or nauseous • Do not take the victim to the hospital yourself Activate EMS! Get an AED if possible to have on stand-byeIf they are able to ingest it and are not allergic to aspirin you should have the victim chew and swallow one adult aspirinSTROKE Strokes can be caused by a blood clot or from a bursting blood vessel in the brain• If someone is having a stroke: activate EMS, stay with the victim and treat for shock, and be prepared to start CPR if needed• Do not give any blood thinning medication such as aspirin to a victim with a possible stroke as they may have bleeding in the brain and this would make it worseThe FAST acronym that can help you identify the warning signs of a stroke is found below: F- facial droop A- arm paralysis", "metadata": {"source": "CPR_AED_and_FIRST_AID_MANUAL.pdf"}}, {"id": "chunk_1", "score": 0.6079697608947754, "content": "BASIC FIRST AID MANUALBASIC FIRST AID MANUALState Disaster Management AuthorityState Disaster Management Authority2829l Call 108 or the local emergency no immediatelyl Have the person stop what he/she is doing and rest comfortably This will ease the heart's need of oxygen Many people experiencing a heart attack find it easier to breathe while sittingl Loosen any tight or uncomfortable clothingl Closely watch the person until advanced medical personnel takes over Notice any changes in the person's appearance or behavior Mentor the person's conditionl Be prepared to perform CPR and use of AED, if available, if the person loses consciousness and stops breathingl Ask the person if he/she has a history of heart disease Some people with heart disease take prescribed medication for chest pain You can help by getting the medication for the person and assisting him/her with taking the prescribed medicationl Offer aspirin, if medically appropriate and locally protocols allow,Adult chain of survival:The key steps to surviving a cardiac arrest in adults are described as the adult chain and each one needs to occur promptly to ensure survival1 Early recognition by a bystander that a problem exists2 Early 108 call to activate the Emergency Medical Services (EMS)3 Early CPR to maintain artificial ventilation and circulation until the EMS arrives4 Early defibrillation to deal with the heart's electrical problems5 Early advanced medical careThe survival rate for cardiac arrest is very low in most countries, including India It is time-critical with the chances of chances of survival decreasing by about 10% for every minute you have to wait for a defibrillator", "metadata": {"source": "Basic_First_Aid_Manual_English.pdf"}}, {"id": "chunk_0", "score": 0.5781621932983398, "content": "delay CPR In practice, these recommendations are very similar Councils place greater emphasis on "never delaying CPR" and for the general public, organizations recommend starting CPR without checking for a pulse Consult local regulators to consider differences in regulation and liability protection for first aid providers Where EMS is available, learners should be encouraged to start compression-only CPR rather than hesitate as they consider the possibility of rescue breaths In some countries, specifically those with a high rate of tuberculosis, rescue breaths may be discouraged Teach first aid providers chest compression-only CPR (or additionally bag-valve-mask resuscitation if they are professional responders) See epidemic and health crises In contexts where there is no EMS or access to advanced care, prepare learners for the likely death of the person who is unresponsive and has abnormal breathing This should include telling them what to do according to local regulations and requirements for registering a death Learner considerations Prioritize training community members most likely to encounter cardiac arrest emergencies in CPR Members include but are not limited to medics, police officers, firefighters and lifeguards Consider that these groups, given their status and role in the community, might make effective educators for the general public (Tweed & Wilson, 1977) Despite advances in resuscitation science and standardized life-support methods, the overall survival rate of out-of-hospital cardiac arrests remains at less than 10% (Bobrow et al, 2010; Kazaure et al, 2013) Build on the motivation that people with family members who are at high risk of cardiac arrest due to illness might have to learn first aid (Huang et al, 2016)Adapt educational tools (eg mannequins and defibrillators), locations (eg waterfront scenarios for lifeguards) and methods to make them accessible and appropriate for learners' needs and abilities (Papalexopoulou et al, 2014; Sopka et al, 2013)These considerations are specific to emergency medical dispatchers Emergency medical dispatchers play a critical role by promptly recognizing cardiac arrest, providing CPR instructions by phone and dispatching EMS with a defibrillator Consider as part of the education for this role: ○ the use of scripted protocols as a helpful way to confirm when a person is in cardiac arrest ○ additional training around the recognition of agonal breathing ○ how to provide CPR instructions for an adult ○ how to provide instructions for both rescue breaths and compressions if the person is a baby or child Dispatchers who communicate using video-assisted emergency calls may need more training for this tool to be effective and widespread within CPR education (Bolle et al, 2008) Bang et al (2000) suggest that dispatchers with further training (eg technical and emotional) are more effective Facilitation tips Emphasize that the chain of survival relies upon: 164|International first aid, resuscitation and education guidelines 2025 ⚙️", "metadata": {"source": "IFRC International First Aid, Resuscitation and Education Guidelines 2025.pdf"}}], "chunks_found": 5, "status": "success"}

- final result: [{"type": "text", "text": "SEARCH_ID: 0\nQUERY: signs of cardiac arrest in elderly\nSPECULATIVE: false\nRESULT:\nRecognizing cardiac arrest in an elderly person (or any adult) is time-critical. The primary signs that a person has suffered a cardiac arrest include:\n\n* **Unresponsiveness:** The person does not respond to verbal commands or physical stimulation (e.g., shaking their shoulders).\n*… [truncated, 2362 chars total]

🔄 [2026-07-09 19:30:44] Subagent update — TEST 2 — Ambiguous Emergency (collapsed grandmother)

Session ID: `81e7a2b1-f674-4d86-8bbc-c95ca527cadd`
Task: `b20eb3a7-005a-401d-a5c3-d15e367a483a`

- tool `search_first_aid_rag` → {"query": "emergency first aid for unresponsive person", "context": "[Source: IFRC International First Aid, Resuscitation and Education Guidelines 2025.pdf | Score: 0.48]\nLearners should practise providing continuous CPR, minimizing any interruptions, while another person sets up the defibrillator and applies the chest pads They should practise resuming CPR immediately after a pretend shock Emphasize that time is critical to a successful outcome for the unresponsive person Evidence indicates survival relies upon: ○ immediate recognition that someone is unresponsive with abnormal breathing ○ early access to help and access to EMS ○ early high-quality CPR ○ early defibrillation with an automated external defibrillator if the machine detects a shockable rhythm Non-professional people can facilitate defibrillator education just as effectively as healthcare professionals (Castrén et al, 2004) Facilitation tools Facilitating defibrillator skills Rapid, good 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standard CPR) ○ early defibrillation with an automated external defibrillator Emphasize the importance of the first aid provider, other bystanders and EMS working together to provide quick and effective care Help learners understand the desired outcomes of CPR: to pump blood around the body (chest compressions) and get oxygen into the lungs (rescue breaths) This keeps the vital organs like the brain alive until defibrillation can take place Define proper compression rate and depth and highlight that the unresponsive person will have the best chance of recovery if chest compressions are of good quality Emphasize that the person should be lying flat on a firm surface if possible Emphasize that starting CPR early has a significant impact on the likelihood of achieving the return of spontaneous circulation for some people experiencing cardiac arrest However, overall, the likelihood of return of spontaneous circulation remains low Ensure learners understand the fundamental components of standard CPR before being taught compression-only CPR (Lam et al, 2007) Facilitation tools Massive multiplayer virtual worlds allow learners to play a role and experience "real-life" scenarios and environments in which to practise their CPR skills If implementing this tool, ensure facilitators understand how to use it and that the technology is not too complex (Creutzfeldt et al, 2013) If video-assisted dispatcher support is used in your country, explain how this might work so that learners are prepared to use it Use role play to help learners to understand what happens when they call an emergency number (Bolle et al, 2008, 2010) If mannequins are unavailable, old car tyres can be used to practise CPR compressions Dig the tyre about two-thirds of the way into the ground to simulate a chest, which will recoil when pushed on For learners who train regularly and are knowledgeable in CPR, extend skills training to include performing CPR in different settings and situations (eg in noisy or distracting environments, with anxious relatives present, in crowds or small spaces with restricted access) Such training exercises encourage team-based approaches and lateral thinking Use film clips or demonstrations to improve learners' recognition of abnormal breathing including agonal breathing and someone who is not breathing Learning connections In contexts where a defibrillator is likely to be available, pair this topic with unresponsive and abnormal breathing when a defibrillator is available Ensure learners also understand how to maintain an open airway if someone is unresponsive and breathing normally The most common condition to result in a person becoming unresponsive with abnormal breathing is a heart attack Conditions that may influence whether rescue breaths could be beneficial include opioid overdose, drowning, choking or remote situations FIRST AID| 165\n\n---\n\n[Source: IFRC International First Aid, Resuscitation and Education Guidelines 2025.pdf | Score: 0.47]\nimprecise due to the small number of participants, the large variability of results and the lack of data A second CEBaP evidence summary compared the recovery position to only doing the jaw thrust, but no studies could be identified Use of the AVPU scale CEBaP updated the evidence summary on the use of the AVPU scale to assess the level of consciousness in adults in 2025 The summary identified SIX diagnostic accuracy studies and one interrater reliability study, comparing the performance of the AVPU scale to the Glasgow Coma Scale (GCS) as a reference test The studies provided evidence of very low certainty indicating that the AVPU scale and the (p)GCS (paediatric GCS) are equally sensitive and specific to identify impaired levels of consciousness in adults and in children Based on one study, both scales also demonstrate similar interrater reliability Given the user-friendliness of the AVPU scale over the (p)GCS, it is conceivable to recommend the use of the AVPU scale to assess the level of consciousness in adults and children As the AVPU scale does not appear to be a sensitive or specific tool for identifying head injuries, it should therefore not be used for this purpose Non-systematic reviews New lateral (side-lying) trauma position for cases of cervical spine injury Hyldmo, Horodyski, Conrad et al (2016) investigated the safety of the new side-lying trauma position in cervical spine injuries in a cadaver model study and found that in the standard recovery position, the range of motion for lateral bending was 119° While both HAINES positions caused a similar range of motion, the new side-lying trauma position resulted in 26° less (P = 0037) The range of motion of the head, neck and upper body in the standard recovery position was 130 mm In comparison, the HAINES positions showed significantly less motion (58 and 46 mm, respectively), while the side-lying trauma position showed even less (40 mm, P = 0067) The authors concluded that in unresponsive trauma 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position while maintaining spinal alignment (strong recommendation, limited evidence) This FIRST AID|157\n\n---\n\n[Source: IFRC International First Aid, Resuscitation and Education Guidelines 2025.pdf | Score: 0.44]\nFacilitation tips Run relevant and realistic scenarios that allow learners to practise their general approach, recognition and first aid skills to care for an unresponsive person who is breathing When facilitating learning about putting someone in the recovery position, focus on the desired outcome of the person (they are on their side and their airway is open) and keep the steps to achieving this as simple as possible Discuss the mechanics of the tongue with regard to keeping an open airway When a person is unresponsive, their muscles relax, which can cause the tongue to block the airway Emphasize the importance of tilting the head back, as that will pull the tongue forward and keep the airway open Moving the person onto their side maintains an open airway as the tongue 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directly to painful stimuli (like squeezing the fingers or pinching the back of the hand) The person may cry, moan or move U Unresponsive The person does not react, either to verbal or painful stimuli For most first aiders, the most important distinction is between responsive and unresponsive, where alert indicates responsiveness and verbal, pain or unresponsive are abnormal, indicating unresponsiveness Learning connections Explore some conditions that could cause unresponsiveness: head injury, stroke, diabetic emergency and seizure FIRST AID|155\n\n---\n\n[Source: IFRC International First Aid, Resuscitation and Education Guidelines 2025.pdf | Score: 0.42]\nDifferentiate between someone feeling faint and someone who is unresponsive and breathing normally Someone who faints should be unresponsive only for a very short amount of time Scientific foundation Systematic reviews Recovery position The International Liaison Committee on Resuscitation (ILCOR) conducted a scoping review in 2020 regarding the recovery position for individuals with a decreased level of consciousness due to non-traumatic causes, who do not require rescue breathing or chest compressions (Singletary et al 2020a, Singletary et al 2020b) The review includes 31 studies, one case report and two letters to the editor It covers a range of participants, including those with reduced responsiveness due to medical conditions (eg stroke), drug overdose or sleep-disordered breathing, as well as healthy individuals, people under medically induced unconsciousness and cadaveric models with spine instability Several recovery positions were studied One study, where a decreased level of responsiveness was the result of an overdose, suggested that lying down in a semi-raised position may be preferable to a side-lying position; however, additional studies need to confirm this finding For the other medical causes of decreased mental status, the side-lying position was associated with beneficial outcomes The studies on sleep-disordered breathing found that side-lying positioning improved apnoea, hypopnoea and oxygen desaturation However, they may not be directly applicable to the use of the recovery position for people with a decreased level of responsiveness from a medical, toxicological or non-traumatic cause An evidence update in 2024 did not reveal any new studies (Greif et al, 2024a; Greif et al, 2024b) The Centre for Evidence-Based Practice (CEBaP) conducted an evidence summary comparing the HAINES (High Arm in Endangered Spine) position to a lateral recovery position The review includes three experimental studies, one study with healthy volunteers and two studies with human cadavers, comparing the HAINES position (with both legs bent at the knee), modified HAINES position (with one leg bent at the knee) and the side-lying trauma position (which requires two rescuers and the use of a cervical collar) to the side-lying recovery position It was shown that the HAINES position resulted in a statistically significant decrease of movement in the cervical region and a decrease of the spinal range of linear motion, compared to the side-lying recovery position However, the HAINES position resulted in a statistically significant increase in movement in the thoracolumbar region, compared to the side-lying recovery position It was shown that the modified HAINES position resulted in a statistically significant decrease in the spinal range of linear motion, compared to the side-lying recovery position The review found that the side-lying trauma position resulted in a statistically significant decrease in the spinal range of angular motion, compared to the side-lying recovery position A statistically significant difference in a range of other motion outcomes could not be demonstrated for any of these alternative positions No other outcomes were measured, and evidence with people with a spine injury is not available Evidence is of very low certainty, and the results of these studies are 156| International first aid, resuscitation and education guidelines 2025 ⚙️", "chunks": [{"id": "chunk_4", "score": 0.47536540031433105, "content": "Learners should practise providing continuous CPR, minimizing any interruptions, while another person sets up the defibrillator and applies the chest pads They should practise resuming CPR immediately after a pretend shock Emphasize that time is critical to a successful outcome for the unresponsive person Evidence indicates survival relies upon: ○ immediate recognition that someone is unresponsive with abnormal breathing ○ early access to help and access to EMS ○ early high-quality CPR ○ early defibrillation with an automated external defibrillator if the machine detects a shockable rhythm Non-professional people can facilitate defibrillator education just as effectively as healthcare professionals (Castrén et al, 2004) Facilitation tools Facilitating defibrillator skills Rapid, good quality CPR remains essential to a successful 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- final result: [{"type": "text", "text": "If you encounter an unresponsive person, follow these emergency steps immediately. Time is critical.\n\n### 1. Assess the Situation\n* **Check for responsiveness:** Gently shake the person's shoulders and shout, "Are you okay?"\n* **Check for breathing:** Look for normal breathing. If the person is not breathing or is breathing abnormally (e.g., gasping), they are … [truncated, 2244 chars total]

🔄 [2026-07-09 19:30:44] Subagent update — TEST 2 — Ambiguous Emergency (collapsed grandmother)

Session ID: `81e7a2b1-f674-4d86-8bbc-c95ca527cadd`
Task: `be3d774c-23ac-4eff-9ad9-0fc3d91485ed`

- tool `generate_alert_report` → 🚨 EMERGENCY ALERT — MedicAI
  Type: Medical Emergency | Severity: Critical
  Patient: Unknown, 72yo, Blood: Unknown
  Symptoms: Collapsed, unresponsive
  Location: 14 Admiralty Way, Lekki Phase 1, Lagos (6.441, 3.456)
- tool `broadcast_to_hospitals` → [{"hospital_id": "hospital_2", "hospital_name": "R-Jolad Hospital", "status": "sent", "channel": "whatsapp", "message_sid": "SMbffc7ae5ea818993645a576106e8390c", "accept_url": "http://localhost:8000/hospital/respond/session_12345_emergency_lekki/hospital_2/accept", "reject_url": "http://localhost:8000/hospital/respond/session_12345_emergency_lekki/hospital_2/reject"}]
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Session ID: `81e7a2b1-f674-4d86-8bbc-c95ca527cadd`
Task: `e18ba33a-67d3-4526-996d-1b119fb17c9f`

- tool `search_first_aid_rag` → {"query": "how to check for breathing in an unconscious person", "context": "[Source: Basic_First_Aid_Manual_English.pdf | Score: 0.52]\nBASIC FIRST AID MANUALBASIC FIRST AID MANUALState Disaster Management AuthorityState Disaster Management Authority1617l Weak or abnormal breath sounds (wheezing, etc)l Occasionally gaspsl Reduced responsivenessl Anxietyl Unusual skin colour (pallor)l Rapid or slow breathingl Unusual postureHow to check for breathing:The rescuer should-l LOOK for movement of the upper abdomen or lower chestl LISTEN for the escape of air from nose and mouthl FEEL for breath on the side of your face / movement of the chest and upper abdomenl THIS SHOULD TAKE YOU NO LONGER THAN 10 SECONDSRescue breaths:Kneel beside the victim’s head Maintained an open airways Use resuscitation barrier devise Take a breath, open your mouth as widely as possible and place it over the victim’s slightly open mouth Whilst maintaining an open airway pinch the nostril (or seal nostrils with rescuer cheeks) and blow to inflate the victim’s lungsBecause the hand supporting the head comes forward some head tilt may be lost and their airway may be obstructed Pulling upwards (with the hand on the chin) helps to reduce this problem For mouth to mouth ventilation, it is reasonable to give each breath in a short time (one second) with a volume to achieve chest rise regardless of the cause of collapse Care should be taken not to over-inflate the chestLook for rise of the victim’s chest whilst inflating If the chest does not rise, possible causes are:l Obstruction in the airway (inadequate head tilt, chin lift, tongue, or foreign body)l Insufficient air-being blown into the lungsl Inadequate air seal around mouth and or noseIf the chest does not rise, ensure correct head tilt, adequate air seal and ventilation Following inflation of the lungs, lift your mouth from your victim’s mouth, turn your head towards the victim’s chest and listen and feel for air being exhaled from the mouth and nose\n\n---\n\n[Source: IFRC International First Aid, Resuscitation and Education Guidelines 2025.pdf | Score: 0.52]\nFirst aid steps If the person’s breathing is abnormal or they are not breathing: 1Immediately shout for help and ask bystanders to access EMS If you are alone, access EMS yourself, either using a telephone with the speaker function or–if there is no phone available nearby–leave the person to access EMS Do this as quickly as possible 2 Begin chest compressions without delay; push down on the centre of the person’s chest at a fast and regular rate (100–120 compressions per minute) 3 If able and willing, combine chest compressions with rescue breaths at a ratio of 30:2 (30 compressions followed by two breaths) Rescue breaths may be particularly beneficial for unresponsive individuals whose condition is related to hypoxia, such as drowning, choking, strangulation or opioid overdose They may also be beneficial when a delay in defibrillation is anticipated Each rescue breath should be delivered over approximately one second, producing visible chest rise The provision of rescue breaths should not interrupt chest compressions for more than 10 seconds If the first aider is unable or unwilling to provide rescue breaths, chest compressions alone should be continued 4 If one is available, ask a bystander to bring an automated external defibrillator as soon as possible Follow the voice prompts, interrupting chest compressions as little as possible5 Continue to give chest compressions and breaths unless otherwise instructed to pause (either by an automated defibrillator or professional responder) Pause compressions if the person shows signs of recovery, such as coughing, opening their eyes, speaking or moving purposefully and breathing normally 6 If two or more first aid providers are present, alternate giving chest compressions and breaths every two minutes to prevent getting tired Ensure that there is no interruption in compressions as the next person

takes over Local adaptation These local adaptations underscore the importance of safety, quality and respect in resuscitation efforts, particularly in challenging or remote scenarios Manual CPR during transport: Delivering manual CPR during transport poses significant risks to providers and can reduce the quality of compressions EMS systems must assess these risks and, when feasible, use strategies to avoid them Resuscitation is best performed at the scene unless there is a clear and justified need for transport If transport is required, EMS providers should prioritize maintaining high-quality CPR throughout the journey Remote area transport: In circumstances where EMS or other advanced care options are unavailable, and transport from a remote location to medical care is necessary, continuous CPR should be performed on a firm surface during transit Dignity and respect: When EMS or advanced care is not accessible, it is critical to uphold the individual's dignity during all aspects of care and transport Drowning and Hypothermia cases: In situations involving drowning or hypothermia, there remains a possibility of a positive response to CPR even when defibrillation is not feasible Continue to perform CPR in these situations for the best possible outcome 162|International first aid, resuscitation and education guidelines 2025 ⚙️\n\n---\n\n[Source: IFRC International First Aid, Resuscitation and Education Guidelines 2025.pdf | Score: 0.51]\nimprecise due to the small number of participants, the large variability of results and the lack of data A second CEBaP evidence summary compared the recovery position to only doing the jaw thrust, but no studies could be identified Use of the AVPU scale CEBaP updated the evidence summary on the use of the AVPU scale to assess the level of consciousness in adults in 2025 The summary identified SIX diagnostic accuracy studies and one interrater reliability study, comparing the performance of the AVPU scale to the Glasgow Coma Scale 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- final result: [{"type": "text", "text": "To check if an unconscious person is breathing, follow these steps. This assessment should take **no longer than 10 seconds**.\n\n### How to Check for Breathing\n1. **LOOK** for movement of the upper abdomen or lower chest.\n2. **LISTEN** for the escape of air from the nose and mouth.\n3. **FEEL** for breath on the side of your face or for movement of the chest and up… [truncated, 1831 chars total]

🔄 [2026-07-09 19:30:44] Subagent update — TEST 2 — Ambiguous Emergency (collapsed grandmother)

Session ID: `81e7a2b1-f674-4d86-8bbc-c95ca527cadd`
Task: `08698dc3-fdbb-4856-9898-ab7405518314`

- tool `search_first_aid_rag` → {"query": "signs and symptoms of stroke in elderly", "context": "[Source: IFRC International First Aid, Resuscitation and Education Guidelines 2025.pdf | Score: 0.62]\nFor the imaging outcome “lesion volume change at six hours, at 24 hours and at hospital discharge”, no difference could be shown in one randomized controlled trial (low-certainty evidence) One observational study also looked at complications and could not show an association between supplementary oxygen on the one hand and pneumonia at hospital discharge, and pulmonary edema and the use of non-invasive positive-pressure ventilation on the other hand, but showed a lower rate of hospital-acquired pneumonia and a higher rate of tracheal intubation and of respiratory complications (very low-certainty evidence) Of the two newly identified randomized controlled trials, one study compared high-flow oxygen with no oxygen but did not find a significant difference in global disability scores The other randomized controlled trial found better outcomes with normobaric hyperoxia compared with room air Since these studies were not yet included in the systematic review, they had not yet been integrated into the Consensus on Science and Treatment Recommendations Education reviews Campaigns to improve public recognition of stroke and instigate fast action to get the person to a medical facility have had mixed results In general, evidence identifies that campaigns based on FAST can increase people’s awareness of the need to act quickly and get the person to emergency medical care following the identification of one or more signs from the FAST mnemonic Wolters et al (2015) retrospectively considered 668 consecutive patients with stroke between 2002–2008 and 2009–2013 In between these time intervals, a national television campaign on stroke recognition and how to act (based on FAST) ran Results showed that patients were statistically more likely to seek medical care within three hours after seeing the campaign The median time to get medical attention fell from 53 to 31 minutes, and arrival at the hospital fell from 185 to 119 minutes post-campaign Robinson et al (2012) surveyed 1,300 people in Leicester (UK) following a campaign on stroke recognition (FAST) and found that participants strongly remembered the campaign and the mnemonic elements After surveying 356 adults in Birmingham (UK) following the same campaign, Beizk (2012) found that 649% were aware of the FAST campaign Of this percentage however, 325% could not recall any letters, 95% remembered one, 130% two, 173% three and 277% all four correctly Wall et al (2008) identified low-quality evidence that training leads to an increase in the ability to identify signs of a stroke After training, those able to identify the signs rose from 764% to 944% Additionally, 969% of participants were able to identify the signs of stroke three months after training Caminiti et al (2013) describe a study to develop a campaign aimed at increasing stroke awareness and preparedness They found that integrating theory with the information collected from target populations enabled them to create effective tools for that audience 374|International first aid, resuscitation and education guidelines 2025 ⚙️\n\n---\n\n[Source: IFRC International First Aid, Resuscitation and

Education Guidelines 2025.pdf | Score: 0.61]\nFor OPSS, there is low-certainty evidence from one observational study with 861 participants suspected of stroke It showed no association between the use of OPSS and the rate of recognition of ischaemic stroke This same study did show an association between the use of OPSS and an increase in the rate of thrombolytic therapy for all people with ischaemic stroke, as well as an association between the use of OPSS and an increased rate of thrombolytic therapy for people with ischaemic stroke arriving within three hours For FASTER, there is very low-certainty evidence from one observational study including 181 participants with suspected acute stroke It showed an association between the use of FASTER and the number of people who received thrombolytic therapy Of patients who had the scale applied, 191% received thrombolytic therapy compared with 75% who did not have the scale applied When looking at studies investigating the correct stroke diagnosis, 19 observational studies were identified enrolling a total of 8,153 people and studying nine different stroke screening assessment systems These studies were divided into subgroups based on whether the stroke scales included a glucose measurement or not Very low-certainty evidence was found on the important outcome of increased public recognition of stroke signs when using a stroke screening assessment system One human study enrolling 72 members of the public measured the difference before training, immediately after, and three months after training The study showed a benefit where 764% of participants (55/72) were able to identify stroke signs before training on how to use a stroke screening assessment system, compared with 944 % (68/72) immediately after training Additionally, 969 % of participants (63/65) were able to identify stroke signs three months after training Supplementary oxygen ILCOR conducted a systematic review of the use of supplementary oxygen for acute 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randomized controlled trial also showed benefit at seven months (low-certainty evidence) For the outcome quality of life, no benefit of supplementary oxygen was shown in two randomized controlled trials, and one trial even showed a lower quality of life (low-certainty evidence) FIRST AID|373\n\n---\n\n[Source: IFRC International First Aid, Resuscitation and Education Guidelines 2025.pdf | Score: 0.59]\nDesign activities that will help learners to: Identify that their job as the first aid provider is to recognize the signs of a stroke and call EMS or get the person to a medical facility (Maze & Bakas, 2004) Develop an understanding of stroke (beyond using FAST) and recognize minor symptoms, such as impaired vision, unsteadiness and headache Develop critical-thinking skills to determine when and how to respond Understand the chain of people involved in providing immediate care to the person experiencing a stroke Engage in the decision-making process of providing care Consider the perceived risks, benefits and barriers to accessing EMS Understand that a quick response time increases the possibility of treatment and recovery (Caminiti et al, 2017) Design activities and training that support the above learning to help first aid providers trust in their ability to act quickly and effectively This will help to reduce delays in care Facilitation tools The intent of the learning activities developed for this topic is to build learners' confidence and connect their understanding of warning signs with actions (Caminiti et al, 2017) These activities are also an opportunity to reinforce positive behaviours Use acronyms such as FAST to help quickly recall sequenced information (Bietzk et al, 2012; Robinson et al, 2012; Wolters et al, 2015) ○ The CPSS assessment tool contains similar physical signs to check (face droop, arm weakness, speech abnormalities) Therefore, you may want to consider adapting the acronym (while maintaining the signs, symptoms and actions to take) into a word that is more suitable in your language and therefore more accessible to your learners Where educators wish to use alternative stroke scale assessment tools such as MASS or LAPSS (because of the likelihood of a first aider being able to take a blood or glucose measurement, for example), ensure that this is both appropriate for your learners and authorized in your country Use multiple resources such as media, community education and professional education when developing the lessons for this topic A layered approach has been shown to successfully reduce prehospital delays (Becker et al, 2001; Caminiti et al, 2017; Flynn et al, 2014; Wall et al, 2008) Stories (either read or performed) and shared personal experiences can support learners' understanding of the topic, especially at low levels of education and literacy (Caminiti et al, 2017)Facilitate a session where learners build case studies and include details such as recognizing the early signs of stroke, accessing EMS and 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the importance of maintaining the person's physical, mental and emotional safety Recovery position: If the person becomes unresponsive, starts drooling or struggles to swallow, the first aid provider should place them on their side with their head tilted back Psychological support: The first aid provider should monitor the person's condition and provide reassurance until medical care arrives A stroke can be an extremely frightening experience, and this is an excellent opportunity to reinforce empathy and build confidence in learners' abilities to provide support to those affected Scientific foundation Systematic reviews Body position The Centre for Evidence-Based Practice (CEBaP) developed an evidence summary on whether the positioning of a person with a stroke affects a variety of outcomes This summary was updated in 2024 and identified one additional studyThere is limited evidence either in favour of lying flat on the back or sitting up A statistically significant increase of a score of 0–2 and 0–6 on the modified Rankin Scale (mRS, a scale indicating the degree of disability) at 90 days could not be demonstrated for either position Additionally, a statistically significant decrease in mortality, recurrent stroke, pneumonia, length of hospital stay and neurological deterioration could not be demonstrated for either position It was shown that lying flat resulted in a statistically significant decrease in mRS 0-2 (disability) and a statistically significant increase in general health score, compared to sitting up However, the guideline developers did not consider these differences to be clinically relevant Evidence is of low certainty and results cannot be considered precise due to limited sample size, low number of events, lack of data and large variability of results Stroke screening assessment systems The International Liaison Committee on Resuscitation (ILCOR) performed a systematic review on the use of stroke assessment systems to aid in the recognition of 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Assessment: While waiting for EMS to arrive or while in transit to medical care, first aid providers should draw from previous learning and recognize the importance of maintaining the person's physical, mental and emotional safety Recovery position: If the person becomes unresponsive, starts drooling or struggles to swallow, the first aid provider should place them on their side with their head tilted back Psychological support: The first aid provider should monitor the person's condition and provide reassurance until medical care arrives A stroke can be an extremely frightening experience, and this is an excellent opportunity to reinforce empathy and build confidence in learners' abilities to provide support to those affected Scientific foundation Systematic reviews Body position The Centre for Evidence-Based Practice (CEBaP) developed an evidence summary on whether the positioning of a person with a stroke affects a variety of outcomes This summary was updated in 2024 and identified 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- final result: [{"type": "text", "text": "SEARCH_ID: stroke_elderly_001\nQUERY: signs and symptoms of stroke in elderly\nSPECULATIVE: false\nRESULT:\nRecognizing the signs of a stroke quickly is critical for improving the chances of treatment and recovery. While the **FAST** mnemonic is the most commonly used tool for identifying the primary signs of a stroke, it is important to be aware of both obvious and more… [truncated, 2133 chars total]

[2026-07-09 19:31:07] TEST 3 — Follow-up: Answer clarifying question

Session ID: `81e7a2b1-f674-4d86-8bbc-c95ca527cadd`
Duration: 24.0s

User message:

> She is not breathing and her lips are turning blue

Main agent — tool calls

1. `resolve_uncertainty` → {"confirmed_search_ids": ["cardiac_arrest_signs"], "cancel_task_ids": ["08698dc3-fdbb-4856-9898-ab7405518314", "d49621bf-9ddb-433b-b42b-182a9e5db69d"], "discard_search_ids": [], "new_certain_queries": ["emergency CPR instructions for cardiac arrest", "how to check for pulse and breathing in unresponsive person"], "summary": "The patient is unresponsive, not breathing, and showing signs of cyanosis… [truncated, 489 chars total]
2. `cancel_async_task` → Cancelled async subagent task: 08698dc3-fdbb-4856-9898-ab7405518314
3. `cancel_async_task` → Cancelled async subagent task: d49621bf-9ddb-433b-b42b-182a9e5db69d
4. `start_async_task` → Launched async subagent. task_id: 2214fd05-3ba8-455b-9e5a-39be82c16486
5. `start_async_task` → Launched async subagent. task_id: 3a767f9d-c818-4f77-b86c-2b6c5e602ff9
6. `check_async_task` → {"status": "running", "thread_id": "2214fd05-3ba8-455b-9e5a-39be82c16486"}
7. `check_async_task` → {"status": "success", "thread_id": "e18ba33a-67d3-4526-996d-1b119fb17c9f", "result": [{"type": "text", "text": "To check if an unconscious person is breathing, follow these steps. This assessment should take **no longer than 10 seconds**.\n\n### How to Check for Breathing\n1. **LOOK** for movement of the upper abdomen or lower chest.\n2. **LISTEN** for the escape of air from the nose and mouth.\… [truncated, 500 chars total]
8. `check_async_task` → {"status": "running", "thread_id": "3a767f9d-c818-4f77-b86c-2b6c5e602ff9"}
9. `check_async_task` → {"status": "success", "thread_id": "b20eb3a7-005a-401d-a5c3-d15e367a483a", "result": [{"type": "text", "text": "If you encounter an unresponsive person, follow these emergency steps immediately. Time is critical.\n\n### 1. Assess the Situation\n* **Check for responsiveness:** Gently shake the person's shoulders and shout, "Are you okay?"\n* **Check for breathing:** Look for normal breathing.… [truncated, 500 chars total]
10. `check_async_task` → {"status": "success", "thread_id": "70a0d01e-191c-4887-9102-b97a29095c20", "result": [{"type": "text", "text": "SEARCH_ID: 0\nQUERY: signs of cardiac arrest in elderly\nSPECULATIVE: false\nRESULT:\nRecognizing cardiac arrest in an elderly person (or any adult) is time-critical. The primary signs that a person has suffered a cardiac arrest include:\n\n* **Unresponsiveness:** The person does not r… [truncated, 500 chars total]
11. `assemble_first_aid_response` → {"priority_steps": ["Call emergency services immediately.", "Place Grace on her back on a firm, flat surface.", "Check for an AED; if available, turn it on and follow the voice prompts.", "Place your hands in the center of her chest.", "Push hard and fast, at least 2 inches deep, at a rate of 100-120 compressions per minute.", "Continue compressions without stopping until help arrives."], "do_not"… [truncated, 500 chars total]
12. `start_async_task` → Launched async subagent. task_id: 27746ca1-a65b-4ca3-9421-08b8b6bbaefb

Subagent progress events (rag / coordinator / video)

- `rag_event` — {"source": "subagent", "task_id": "d49621bf-9ddb-433b-b42b-182a9e5db69d", "status": "complete", "tool_calls": [{"name": "search_first_aid_rag", "content": "{"query": "causes of sudden collapse

in elderly", "context": "[Source: CPR_AED_and_FIRS… [truncated, 18352 chars total]
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Subagent internals (fetched from subagent server)

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  - tool `search_first_aid_rag` → {"query": "CPR steps for adult cardiac arrest", "context": "[Source: IFRC International First Aid, Resuscitation and Education Guidelines 2025.pdf | Score: 0.49]\nAccess help When speaking to EMS, very clearly explain that the unresponsive person is not breathing normally; this will prompt EMS to prioritize your case appropriately Ask bystanders for help accessing EMS and providing CPR, as well as bringing and using an automated external defibrillator An unresponsive person who is not breathing is unlikely to achieve spontaneous circulation from CPR alone Their heart needs an electric shock from a defibrillator It is vital that EMS arrive and a defibrillator is used The survival of the person depends upon immediate and effective CPR; when accessing help, keep any interruptions to chest compressions minimal Recovery Even if the first aid provider has performed CPR and defibrillation and the person is now responsive and breathing normally, you must continue close monitoring until EMS arrives as the person may stop breathing again Education considerations Supporting learners to have the confidence and willingness to attempt CPR on a person who is unresponsive and breathing abnormally is a priority for first aid educators Remember that the opportunity to provide CPR is generally very low—some learners might never have to perform it However, if such a situation does arise, learners need to be prepared for feelings of doubt, uncertainty and lack of confidence Therefore, CPR education should always take these feelings into account, and support the learner to overcome them Context considerations Refer to and follow the guidance of regional resuscitation councils or other national protocols and tailor education accordingly ○ Recognizing cardiac arrest: ■ Councils emphasize the recognition of abnormal breathing and the possibility of brief convulsions or emphasize the absence or abnormality of breathing in an unconscious person It is an almost identical approach: some councils specify the possibility of brief convulsions that may delay recognition of cardiac arrest For the general public, all

organizations recommend starting CPR without checking for a pulse ○ Activating emergency services (EMS): ■ Councils recommend calling emergency services before assessing breathing if the person is unconscious or recommend calling emergency services after identifying cardiac arrest (unconsciousness and abnormal or no ventilation) These approaches are practically identical, but the councils place greater emphasis on obtaining assistance from an EMS dispatcher as quickly as possible This recommendation must be implemented according to the availability of the EMS system ○ Firm surface for compressions ■ For some councils, immediate CPR is recommended, even on a bed, to avoid no-flow because lifting the victim delays treatment; other councils recommend a hard surface, but also state that moving the victim should not FIRST AID|163\n\n---\n\n[Source: IFRC International First Aid, Resuscitation and Education Guidelines 2025.pdf | Score: 0.48]\ndelay CPR In practice, these recommendations are very similar Councils place greater emphasis on "never delaying CPR" and for the general public, organizations recommend starting CPR without checking for a pulse Consult local regulators to consider differences in regulation and liability protection for first aid providers Where EMS is available, learners should be encouraged to start compression-only CPR rather than hesitate as they consider the possibility of rescue breaths In some countries, specifically those with a high rate of tuberculosis, rescue breaths may be discouraged Teach first aid providers chest compression-only CPR (or additionally bag-valve-mask resuscitation if they are professional responders) See epidemic and health crises In contexts where there is no EMS or access to advanced care, prepare learners for the likely death of the person who is unresponsive and has abnormal breathing This should include telling them what to do according to local regulations and requirements for registering a death Learner 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⚙️\n\n---\n\n[Source: IFRC International First Aid, Resuscitation and Education Guidelines 2025.pdf | Score: 0.48]\n○ immediate recognition that someone is unresponsive and breathing abnormally ○ early access to help and EMS ○ early high-quality CPR (compression-only or standard CPR) ○ early defibrillation with an automated external defibrillator Emphasize the importance of the first aid provider, other bystanders and EMS working together to provide quick and effective care Help learners understand the desired outcomes of CPR: to pump blood around the body (chest compressions) and get oxygen into the lungs (rescue breaths) This keeps the vital organs like the brain alive until defibrillation can take place Define proper compression rate and depth and highlight that the unresponsive person will have the best chance of recovery if chest compressions are of good quality Emphasize that the person should be lying flat on a firm surface if possible Emphasize that starting CPR early has a 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attack Conditions that may influence whether rescue breaths could be beneficial include opioid overdose, drowning, choking or remote situations FIRST AID|165\n\n---\n\n[Source: cpr-aed-first-aid-handbook-disque-obooko.pdf | Score: 0.47]\n28ADULT CPRCPR is comprised of chest compressions, airway management, and rescue breathing To deliverhigh-quality CPR, you must begin high-quality chest compressions quickly, as these are consideredthe most important factor in giving the person a chance to recover Compressing the chestcirculates blood to the brain and the heart High-quality chest compressions are delivered at a ratebetween 100 to 120 beats per minute and at a depth between 2 to 24 inches (5 to 6 cm)The importance of early initiation of CPR by lay rescuers has been re-emphasized by the 2025ILCOR Guidelines for CPR The risk of harm to the patient is low if the patient is not in cardiacarrest Bystanders should not be afraid to start CPR even if they are not sure whether the victim 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CPR, AEDAND CHOKING 5CPR – CPR, AED & First Aid", "chunks": [{"id": "chunk_4", "score": 0.4859127402305603, "content": "Access help When speaking to EMS, very clearly explain that the unresponsive person is not breathing normally; this will prompt EMS to prioritize your case appropriately Ask bystanders for help accessing EMS and providing CPR, as well as bringing and using an automated external defibrillator An unresponsive person who is not breathing is unlikely to achieve spontaneous circulation from CPR alone Their heart needs an electric shock from a defibrillator It is vital that EMS arrive and a defibrillator is used The survival of the person depends upon immediate and effective CPR; when accessing help, keep any interruptions to chest compressions minimal Recovery Even if the first aid provider has performed CPR and defibrillation and the person is now responsive and breathing normally, you must continue close monitoring until EMS arrives as the person may stop breathing again 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Education Guidelines 2025.pdf"}}, {"id": "chunk_3", "score": 0.48443201184272766, "content": "delay CPR In practice, these recommendations are very similar Councils place greater emphasis on "never delaying CPR" and for the general public, organizations recommend starting CPR without checking for a pulse Consult local regulators to consider differences in regulation and liability protection for first aid providers Where EMS is available, learners should be encouraged to start compression-only CPR rather than hesitate as they consider the possibility of rescue breaths In some countries, specifically those with a high rate of tuberculosis, rescue breaths may be discouraged Teach first aid providers chest compression-only CPR (or additionally bag-valve-mask resuscitation if they are professional responders) See epidemic and health crises In contexts where there is no EMS or access to advanced care, prepare learners for the likely death of the person who is unresponsive and has abnormal 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  - tool `search_first_aid_rag` → {"query": "how to check for pulse and breathing in unresponsive person", "context": "[Source: IFRC International First Aid, Resuscitation and Education Guidelines 2025.pdf | Score: 0.52]\n○ immediate recognition that someone is unresponsive and breathing abnormally ○ early access to help and EMS ○ early high-quality CPR (compression-only or standard CPR) ○ early defibrillation with an automated external defibrillator Emphasize the importance of the first aid provider, other bystanders and EMS working together to provide quick and effective care Help learners understand the desired outcomes of CPR: to pump blood around the body (chest compressions) and get oxygen into the lungs (rescue breaths) This keeps the vital organs like the brain alive until defibrillation can take place Define proper compression rate and depth and highlight that the unresponsive person will have the best chance of recovery if chest compressions are of good quality Emphasize that the person should be lying flat on a firm surface if possible Emphasize that

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abnormal breathing is a heart attack Conditions that may influence whether rescue breaths could be beneficial include opioid overdose, drowning, choking or remote situations FIRST AID|165\n\n---\n\n[Source: IFRC International First Aid, Resuscitation and Education Guidelines 2025.pdf | Score: 0.51]\ndelay CPR In practice, these recommendations are very similar Councils place greater emphasis on "never delaying CPR" and for the general public, organizations recommend starting CPR without checking for a pulse Consult local regulators to consider differences in regulation and liability protection for first aid providers Where EMS is available, learners should be encouraged to start compression-only CPR rather than hesitate as they consider the possibility of rescue breaths In some countries, specifically those with a high rate of tuberculosis, rescue breaths may be discouraged Teach first aid providers chest compression-only CPR (or additionally bag-valve-mask resuscitation if they are 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learning about putting someone in the recovery position, focus on the desired outcome of the person (they are on their side and their airway is open) and keep the steps to achieving this as simple as possible Discuss the mechanics of the tongue with regard to keeping an open airway When a person is unresponsive, their muscles relax, which can cause the tongue to block the airway Emphasize the importance of tilting the head back, as that will pull the tongue forward and keep the airway open Moving the person onto their side maintains an open airway as the tongue will fall forward and any blood or vomit can drain out Identify that a person can suddenly become unresponsive (due to a stroke, electrocution, head injury, etc) or gradually (by certain poisonings or a diabetic emergency) Identify that while an altered mental status is frequently a separate issue, the person may display signs before becoming unresponsive First aid providers may be able to intervene before the person becomes 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Explore some conditions that could cause unresponsiveness: head injury, stroke, diabetic emergency and seizure FIRST AID|155\n\n---\n\n[Source: IFRC International First Aid, Resuscitation and Education Guidelines 2025.pdf | Score: 0.46]\nDifferentiate between someone feeling faint and someone who is unresponsive and breathing normally Someone who faints should be unresponsive only for a very short amount of time Scientific foundation Systematic reviews Recovery position The International Liaison Committee on Resuscitation (ILCOR) conducted a scoping review in 2020 regarding the recovery position for individuals with a decreased level of consciousness due to non-traumatic causes, who do not require rescue breathing or chest compressions (Singletary et al 2020a, Singletary et al 2020b) The review includes 31 studies, one case report and two letters to the editor It covers a range of participants, including those with reduced responsiveness due to medical conditions (eg stroke), drug overdose or sleep-disordered breathing, as well as healthy individuals, people under medically induced unconsciousness and cadaveric models with spine instability Several recovery positions were studied One study, where a decreased level of responsiveness was the result of an overdose, suggested that lying down in a semi-raised position may be preferable to a side-lying position; however, additional studies need to confirm this finding For the other medical causes of decreased mental status, the side-lying position was associated with beneficial outcomes The studies on sleep-disordered breathing found that side-lying positioning improved apnoea, hypopnoea and oxygen desaturation However, they may not be directly applicable to the use of the recovery position for people with a decreased level of responsiveness from a medical, toxicological or non-traumatic cause An evidence update in 2024 did not reveal any new studies (Greif et al, 2024a; Greif et al, 2024b) The Centre for 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first aid provider, other bystanders and EMS working together to provide quick and effective care Help learners understand the desired outcomes of CPR: to pump blood around the body (chest compressions) and get oxygen into the lungs (rescue breaths) This keeps the vital organs like the brain alive until defibrillation can take place Define proper compression rate and depth and highlight that the unresponsive person will have the best chance of recovery if chest compressions are of good quality Emphasize that the person should be lying flat on a firm surface if possible Emphasize that starting CPR early has a significant impact on the likelihood of achieving the return of spontaneous circulation for some people experiencing cardiac arrest However, overall, the likelihood of return of spontaneous circulation remains low Ensure learners understand the fundamental components of standard CPR before being taught compression-only CPR (Lam et al, 2007) Facilitation tools Massive multiplayer virtual worlds allow learners to play a role and experience “real-life” scenarios and environments in which to practise their CPR skills If implementing this tool, ensure facilitators understand how to use it and that the technology is not too complex (Creutzfeldt et al, 2013) If video-assisted dispatcher support is used in your country, explain how this might work so that learners are prepared to use it Use role play to help learners to understand what happens when they call an emergency number (Bolle et al, 2008, 2010) If mannequins are unavailable, old car tyres can be used to practise CPR compressions Dig the tyre about two-thirds of the way into the ground to simulate a chest, which will recoil when pushed on For learners who train regularly and are knowledgeable in CPR, extend skills training to include performing CPR in different settings and situations (eg in noisy or distracting environments, with anxious relatives present, in crowds or small spaces with restricted access) Such training exercises encourage team-based approaches and lateral thinking Use film clips or demonstrations to improve learners’ recognition of abnormal breathing including agonal breathing and someone who is not breathing Learning connections In contexts where a defibrillator is likely to be available, pair this topic with unresponsive and abnormal breathing when a defibrillator is available Ensure learners also understand how to maintain an open airway if someone is unresponsive and breathing normally The most common condition to result in a person becoming unresponsive with abnormal breathing is a heart attack Conditions that may influence whether rescue breaths could be beneficial include opioid overdose, drowning, choking or remote situations FIRST AID|165", "metadata": {"source": "IFRC International First Aid, Resuscitation and Education Guidelines 2025.pdf"}}, {"id": "chunk_3", "score": 0.5132114887237549, "content": "delay CPR In practice, these recommendations are very similar Councils place greater emphasis on "never delaying CPR" and for the general public, organizations recommend starting CPR without checking for a pulse Consult local regulators to consider differences in regulation and liability protection for first aid providers Where EMS is available, learners should be encouraged to start compression-only CPR rather than hesitate as they consider the possibility of rescue breaths In some countries, specifically those with a high rate of tuberculosis, rescue breaths may be discouraged Teach first aid providers chest compression-only CPR (or additionally bag-valve-mask resuscitation if they are professional responders) See epidemic and health crises In contexts where there is no EMS or access to advanced care, prepare learners for the likely death of the person who is unresponsive and has abnormal breathing This should include telling them what to do according to local regulations and requirements for registering a death Learner considerations

Prioritize training community members most likely to encounter cardiac arrest emergencies in CPR Members include but are not limited to medics, police officers, firefighters and lifeguards Consider that these groups, given their status and role in the community, might make effective educators for the general public (Tweed & Wilson, 1977) Despite advances in resuscitation science and standardized life-support methods, the overall survival rate of out-of-hospital cardiac arrests remains at less than 10% (Bobrow et al, 2010; Kazaure et al, 2013) Build on the motivation that people with family members who are at high risk of cardiac arrest due to illness might have to learn first aid (Huang et al, 2016)Adapt educational tools (eg mannequins and defibrillators), locations (eg waterfront scenarios for lifeguards) and methods to make them accessible and appropriate for learners' needs and abilities (Papalexopoulou et al, 2014; Sopka et al, 2013)These considerations are specific to emergency medical dispatchers Emergency medical dispatchers play a critical role by promptly recognizing cardiac arrest, providing CPR instructions by phone and dispatching EMS with a defibrillator Consider as part of the education for this role: ○ the use of scripted protocols as a helpful way to confirm when a person is in cardiac arrest ○ additional training around the recognition of agonal breathing ○ how to provide CPR instructions for an adult ○ how to provide instructions for both rescue breaths and compressions if the person is a baby or child Dispatchers who communicate using video-assisted emergency calls may need more training for this tool to be effective and widespread within CPR education (Bolle et al, 2008) Bang et al (2000) suggest that dispatchers with further training (eg technical and emotional) are more effective Facilitation tips Emphasize that the chain of survival relies upon: 164|International first aid, resuscitation and education guidelines 2025 ⚙️", "metadata": {"source": "IFRC International First Aid, Resuscitation and Education Guidelines 2025.pdf"}}, {"id": "chunk_2", "score": 0.4854068160057068, "content": "First aid steps If the person's breathing is abnormal or they are not breathing: 1Immediately shout for help and ask bystanders to access EMS If you are alone, access EMS yourself, either using a telephone with the speaker function or–if there is no phone available nearby–leave the person to access EMS Do this as quickly as possible 2 Begin chest compressions without delay; push down on the centre of the person's chest at a fast and regular rate (100–120 compressions per minute) 3 If able and willing, combine chest compressions with rescue breaths at a ratio of 30:2 (30 compressions followed by two breaths) Rescue breaths may be particularly beneficial for unresponsive individuals whose condition is related to hypoxia, such as drowning, choking, strangulation or opioid overdose They may also be beneficial when a delay in defibrillation is anticipated Each rescue breath should be delivered over approximately one second, producing visible chest rise The provision of rescue breaths should not interrupt chest compressions for more than 10 seconds If the first aider is unable or unwilling to provide rescue breaths, chest compressions alone should be continued 4 If one is available, ask a bystander to bring an automated external defibrillator as soon as possible Follow the voice prompts, interrupting chest compressions as little as possible5 Continue to give chest compressions and breaths unless otherwise instructed to pause (either by an automated defibrillator or professional responder) Pause compressions if the person shows signs of recovery, such as coughing, opening their eyes, speaking or moving purposefully and breathing normally 6 If two or more first aid providers are present, alternate giving chest compressions and breaths every two minutes to prevent getting tired Ensure that there is no interruption in compressions as the next person takes over Local adaptation These local adaptations underscore the importance of safety, quality and respect in resuscitation efforts, particularly in challenging or remote scenarios Manual CPR during transport: Delivering manual CPR during transport poses significant risks to providers and can reduce the quality of compressions EMS systems must assess these risks and, when feasible, use strategies to avoid them Resuscitation is best performed at the scene unless there is a clear and justified need for transport If transport is required, EMS providers should prioritize maintaining high-quality CPR throughout the journey Remote area transport: In circumstances where EMS or other advanced care options are unavailable,

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"
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", "metadata": {"source": "IFRC International First Aid, Resuscitation and Education Guidelines 2025.pdf"}}, {"id": "chunk_0", "score": 0.4626978635787964, "content": "
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  - final result: [{"type": "text", "text": "SEARCH_ID: check_pulse_breathing_unresponsive\nQUERY: how to check for pulse and breathing in unresponsive person\nSPECULATIVE: false\nRESULT:\nWhen encountering an unresponsive person, the primary focus is on assessing their breathing to determine if they require CPR.\n\n### Key Guidelines for Assessment:\n* **Do Not Check for a Pulse:** For the general public, curren… [truncated, 2248 chars total]

- **task** `27746ca1-a65b-4ca3-9421-08b8b6bbaefb`
  - tool `search_youtube` → [{"title": "First Aid Training: CPR", "url": "https://www.youtube.com/watch?v=TsJ49Np3HS0", "thumbnail": "https://i.ytimg.com/vi/TsJ49Np3HS0/mqdefault.jpg", "channel": "St John Ambulance", "description": "Learn the procedure for performing CPR on an unresponsive person in this video from St John Ambulance. CPR stands for ..."}, {"title": "🚨Learn how to do CPR on an adult in the case of a Sudden Cardiac Arrest #viral #cpr #howto", "url": "https://www.youtube.com/watch?v=_F4Of33ifbw", "thumbnail": "https://i.ytimg.com/vi/_F4Of33ifbw/mqdefault.jpg", "channel": "St John Victoria", "description": ""}, {"title": "How to do CPR on an Adult - First Aid Training - St John Ambulance", "url": "https://www.youtube.com/watch?v=BQNNOh8c8ks", "thumbnail": "https://i.ytimg.com/vi/BQNNOh8c8ks/mqdefault.jpg", "channel": "St John Ambulance", "description": "In the event of a pandemic, do not put your face near the casualty's. Instead, look for the chest rising and falling to check for ..."}, {"title": "How To Perform CPR: A Step By Step Guide | Resuscitation CPR First Aid Training", "url": "https://www.youtube.com/watch?v=ewdKM9NYo1A", "thumbnail": "https://i.ytimg.com/vi/ewdKM9NYo1A/mqdefault.jpg", "channel": "Get Licensed - Frontline Security", "description": "Welcome to our life-saving tutorial on "How to do CPR: A Step By Step Guide"! In this comprehensive video, we will take you ..."}, {"title": "How to give hands-only CPR. #HowTo #CPRSteps", "url": "https://www.youtube.com/watch?v=IQAnNRQ8W_g", "thumbnail": "https://i.ytimg.com/vi/IQAnNRQ8W_g/mqdefault.jpg", "channel": "American Red Cross", "description": "This is how to give hands-only CPR to someone following a cardiac emergency. Learning how to do CPR can help save ..."}]

- final result: [{"type": "text", "text": "VIDEOS_READY: [{"channel": "St John Ambulance", "description": "Learn the procedure for performing CPR on an unresponsive person in this video from St John Ambulance. CPR stands for ...", "thumbnail": "https://i.ytimg.com/vi/TsJ49Np3HS0/mqdefault.jpg\", "title": "First Aid Training: CPR", "url": "https://www.youtube.com/watch?v=TsJ49Np3HS0\"}, {"channe… [truncated, 2005 chars total]

Final response to user

> *(empty response)*

Response length: 0 chars

[2026-07-09 19:32:13] TEST 4 — Follow-up: Ask about CPR technique

Session ID: `81e7a2b1-f674-4d86-8bbc-c95ca527cadd`
Duration: 5.4s

User message:

> Can you explain exactly how to do chest compressions? I've never done CPR before

Main agent — tool calls

1. `check_async_task` → {"status": "success", "thread_id": "2214fd05-3ba8-455b-9e5a-39be82c16486", "result": [{"type": "text", "text": "If you encounter someone who is unresponsive and not breathing (or only gasping), follow these emergency steps immediately:\n\n### 1. Ensure Safety and Assess\n* **Check the scene:** Make sure the area is safe for you to approach.\n* **Check responsiveness:** Tap the person firmly an… [truncated, 500 chars total]

Subagent progress events (rag / coordinator / video)

- `rag_event` — {"source": "subagent", "task_id": "d49621bf-9ddb-433b-b42b-182a9e5db69d", "status": "complete", "tool_calls": [{"name": "search_first_aid_rag", "content": "{"query": "causes of sudden collapse in elderly", "context": "[Source: CPR_AED_and_FIRS… [truncated, 18352 chars total]
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- `rag_event` — {"source": "subagent", "task_id": "2214fd05-3ba8-455b-9e5a-39be82c16486", "status": "complete", "tool_calls": [{"name": "search_first_aid_rag", "content": "{"query": "CPR steps for adult cardiac arrest", "context": "[Source: IFRC International… [truncated, 32087 chars total]
- `rag_event` — {"source": "subagent", "task_id": "3a767f9d-c818-4f77-b86c-2b6c5e602ff9", "status": "complete", "tool_calls": [{"name": "search_first_aid_rag", "content": "{"query": "how to check for pulse and breathing in unresponsive person", "context": "[S… [truncated, 34729 chars total]
- `rag_event` — {"source": "subagent", "task_id": "27746ca1-a65b-4ca3-9421-08b8b6bbaefb", "status": "complete", "tool_calls": [{"name": "search_youtube", "content": "[{"title": "First Aid Training: CPR", "url": "https://www.youtube.com/watch?v=TsJ49Np3HS0\", … [truncated, 4028 chars total]

Subagent internals (fetched from subagent server)

no subagent tasks launched

Final response to user

> *(empty response)*

Response length: 0 chars
